



The mixed fleet vehicle routing problem with partial recharging by multiple chargers: Mathematical model and adaptive large neighborhood search

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ABSTRACT

We introduce the Mixed Fleet Vehicle Routing Problem with Time Windows and Partial Recharging by Multiple Chargers (MF-VRP-MC). It composes of electric and internal combustion vehicles, and consolidates several aspects in a comprehensive unique model. The MF-VRP-MC considers travelled distance and carried load on vehicles in both emission and energy consumption functions. It deals with the minimization of total cost while satisfying customer delivery demands. First, we develop a mixed integer mathematical programming formulation for the MF-VRP-MC. Because of the NP-hardness of the problem, to solve medium and large-size instances, then we develop an Adaptive Large Neighborhood Search (ALNS) based algorithm with introducing new advanced neighborhood mechanisms to successfully handle complex problem constraints. Meantime, new approaches are tailored for boosting diversification effect in addition to new neighborhood scoring policy and new enhancement procedure. Furthermore, selection of recharging technology among the others at charging station is firstly considered in the solution phase of a mixed fleet problem. Extensive computational results indicate that our ALNS performs quite well on benchmark instances.

1. Introduction

Worldwide concepts of green logistics include all efforts to minimize or completely eliminate the detrimental effects of real-world logistics applications by measuring the ecological reflections of logistics activities. Transportation has a major share in the global energy consumption and emissions, especially by the vehicles that consume fossil fuels. For this reason, some measures and regulations are applied by governments to reduce emissions and dependence on fossil fuels.

Green routing problem has received great attention from the scientific community due to its practical importance. Green routing problems concentrates on environmental protection from different perspectives such as:

- Minimizing greenhouse gas emissions caused by fuel consumption of internal combustion vehicles (ICVs) (Figliozzi, 2010; Zhang et al., 2015; Li et al., 2018; Olgun et al., 2021; Dewi and Utama, 2021),
- Using alternative fuel vehicles (AFVs) instead of (ICVs) (Erdoğan and Miller-Hooks, 2012; Montoya et al., 2016; Bruglieri et al., 2019; Sadati and Çatay, 2021; Bruglieri, 2022),
- Using electric vehicles (EVs) or hybrid electric vehicles (HEVs) instead of (ICVs) (Felipe et al., 2014; Schneider et al., 2014; Keskin and Çatay, 2016; Montoya et al., 2017; Pelletier et al., 2017; Keskin and Çatay, 2018; Guo et al., 2022),

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Additionally, researchers provide solutions to green routing problems by means of both exact algorithms (Desaulniers et al., 2016; Andelmin and Bartolini, 2017; Leggieri and Haouari, 2017; Lam et al., 2022), heuristics (Rezgui et al., 2019; Olgun et al., 2021; Dewi and Utama, 2021; Bruglieri, 2022) and hybrid algorithms (Koç and Karaoğlu, 2016; Keskin and Çatay, 2018; Hiermann et al., 2019). We also refer the interested readers to review papers of Ghorbani et al. (2020), Marrekchi et al. (2021) and Asghari et al. (2021) on green routing problems.

Electric vehicle (EV) technology, which does not cause emissions while driving if the electricity production process is ignored, also provides high energy efficiency, operates with low noise and necessitates lower maintenance costs (Arslan et al., 2015). In parallel with that emerging, efficient and green technology, its integration with vehicle routing problems (VRPs) has received great attention in literature (see Cortés-Murcia et al. 2019; Vidal et al., 2019; Tahami et al., 2020; Küçükoğlu et al., 2021). This integration effort accompanies additional complexity which originates from the need of relatively long recharging process due to the EV lack of long autonomy.

1.1. Literature review

In addition to the criteria that cause VRPs to be classified among the others in the literature, it is observed that electrical VRPs (E-VRPs) also differ from each other within the framework of the following features. (Erdelić and Carić, 2019; Vidal et al., 2019; Xiao et al., 2021).

Charging configuration types: In some of E-VRPs, uniform charging stations (CSs) with same features are considered. While in some problems, CSs with different configurations, such as normal, fast and super-fast, that control the battery charging speed are considered (Felipe et al., 2014; Keskin and Çatay, 2018). In this case, faster charging capability will be more costly. On the other hand, time related constraints may force faster configurations. Some problems concentrate on the replacement, i.e. swap, of a low or discharged battery with a fully charged battery at the CS (Yang and Sun, 2015; Chen et al., 2021). Moreover two energy supply technologies, both plug-in and battery swapping may be selected in a single CS as in the paper of (Paz et al., 2018). Furthermore, there emerges advanced technologies such as regenerative breaking or wireless recharging.

Charging functions: Some E-VRPs consider the charge level as a time-dependent linear function during the charging process, while the others make use of a non-linear charge function created by observing real data and parameters (Montoya et al., 2017; Pelletier et al., 2017; Koç et al., 2019; Guo et al., 2022). Nonlinear function is converted to piecewise linear approximation where solving the problem becomes more difficult by the effect of breakpoints in the function and the varying rate of state of charge (SoC) during charging.

Charging policies: Some of E-VRPs assume charging the battery up to full capacity each time at the recharging stations, while the others require partial recharging to a reasonable and needed level not to incur redundant costs. It is exerted that partial recharging policy may cause more effective solutions in terms of cost and energy in real life applications (Felipe et al., 2014; Keskin and Çatay, 2016; Keskin and Çatay, 2018; Schiffer and Wallter, 2018).

Engine configurations: Most of E-VRPs focus on fleet consisting of only EVs. Several papers addressed mixed fleet composed of both EVs and internal combustion vehicles (ICVs) (Gonçalves et al., 2011; Sassi et al., 2014; Goeke and Schneider, 2015; Macrina et al., 2019a; Macrina et al., 2019b; Vincent et al., 2021).

Charge consumption functions: A number of E-VRPs assume that the battery charge level will decrease linearly depending on only the distance traveled within the planned routes, while others use on complicated charge consumption models comprehensively reflecting real-life aspects such as speed, acceleration, vehicle weight, road gradient and rolling resistance (Goeke and Schneider, 2015; Lin et al., 2016; Macrina et al. 2019b; Vincent et al., 2021);

Vehicles in the fleet of a transportation company can benefit from CSs of other companies (Arslan et al., 2019; Koç et al., 2019). The E-VRP has been extended to a more holistic perspective such as cost-effective location of CS and optimizing the route of vehicles by considering the predeployment of stations which is known as location routing problem (Yang and Sun, 2015; Schiffer and Wallter, 2017, 2018; Koç et al., 2019; Arslan, 2021; Chen et al., 2021; Guo et al., 2022).

It is also possible to observe some uncertainties in real life applications. For instance, recharging station may not always be available due to the use of different customers. In this case, the parameters affecting the station availability such as the inter-arrival time of the vehicles to the station and the average recharging time of the vehicles at the station can be defined by using probabilistic distributions. Additionally, the travel time of a vehicle between two nodes can be considered stochastic since it can be dependent on traffic density, accidents, roadwork probabilities on the route, and weather conditions in addition to distance. Moreover, demands may also be stochastic. Some of recent stochastic E-VRPs are studied by Rabbani et al. (2018) and Kumar et al. (2022).

Here we concentrate on E-VRPs with the mixed fleet in order to benchmark our study. One of the first studies on mixed fleet E-VRP is addressed by Gonçalves et al. (2011). They studied pickup and delivery version consisting of EVs and ICVs and considered a real-life application of battery distributor in Portugal. Vehicles can be recharged at any time on the route. Each vehicle has a fixed travelling range and a fixed recharging time. Sassi et al. (2014) studied the heterogeneous E-VRP with time dependent charging costs and mixed fleet. EVs are heterogeneous due to differences in battery capacities and operating costs. The unit recharging cost at stations varies depending on the technology. Furthermore, partial recharging policy is permitted. Goeke and Schneider (2015) studied the E-VRP with time windows and mixed fleet. The charge consumption function is modelled as being affected by speed, weight and road gradient. The paper proposed an adaptive large neighborhood search (ALNS) to solve the problem. Macrina et al. (2019a) studied mixed fleet VRP with partial battery charging and time windows. The authors proposed an iterative local search heuristic. They considered an upper bound for the amount of emissions which depends on the carried load in addition to travelled distance. Multiple technologies (single technology per CS) are also integrated into the mathematical model but in the experiments, it is assumed that all CSs have an identical

technology infrastructure. Later on, [Macrina et al. \(2019b\)](#) investigated green VRP with mixed vehicle fleet by introducing a comprehensive energy consumption model. It depends on speed, acceleration, deceleration, load cargo and gradients. However, limiting amount of emission is disregarded in this paper. Both [Macrina et al. \(2019a\)](#) and [Macrina et al. \(2019b\)](#) considered only one type of charging configuration in their experiments. Selecting one of the charging configuration types whose charging rate and cost vary at a single CS is not a decision point in context of their optimization. [Hiemann et al. \(2019\)](#) investigated a mixed fleet problem composed of ICVs, HEVs and EVs. In this study, EVs require recharging while HEVs can optionally switch engine configuration to fossil fuel so as not to visit CS for recharging. Fast refueling operations of ICVs are ignored in the problem. They proposed a metaheuristic combining genetic algorithm with local search and large neighborhood search and finally with integer programming solver. [Al-dal'ain and Çelebi \(2021\)](#) proposed an integrated model for a mixed fleet of EVs and ICVs considering both two dependent decisions which are routing and fleet composition. They firstly optimize the routes for different fleet compositions. Then, results from different compositions are integrated into the replacement model as an input. The refueling of ICVs and recharging of EVs are performed at the depot up to the maximum capacity. Thus, recharging and refueling decisions are out of the scope. Most recently, [Vincent et al. \(2021\)](#) studied a mixed fleet VRP with realistic energy consumption model. Total emitted pollution cost is included in the objective function. They used an ALNS to solve the problem.

In this paper, our motivation is based on the progressive transition to EVs in transportation sector. Several aspects such as mixed fleet, partial recharge policy, selection of recharging technology, emission and energy consumption functions depending on not only travelled distance but also load, emission limitation are consolidated in a comprehensive unique model. To our knowledge, this is the first study that describes the problem from such a holistic view. [Table 1](#) presents the similarities and differences between the most similar mixed fleet problems briefly explained above and our problem.

As shown in [Table 1](#), the main differences between our work and the previous ones are modeling the problem without dummy nodes and selecting charging technology at CSs. We prefer the formulation without dummy nodes based on our preliminary computational analyses. Since this formulation does not augment the network unlike the other formulation which is copying the CSs for multiple visits in the literature, it can be capable to solve larger-.

size problems optimally. Furthermore, the selection of recharging technology among the others at visited CS is considered in the solution phase of a mixed fleet problem. Therefore, which type of charging configuration (normal, fast, super-fast, etc.) to use at CS visited for recharging becomes another challenging question to be answered.

1.2. Contributions and flow of the paper

We make three scientific contributions. We first introduce the Mixed Fleet Vehicle Routing Problem with Time Windows and Partial Recharging by Multiple Chargers (MF-VRP-MC) as a new practical as well as complex VRP variant. We consider load in both energy consumption function for EVs and emission function for ICVs. We limit the emission amounts of ICVs through an upper bound to reflect green motivation in practice. CSs include several recharging technologies whose unit recharging speed and in parallel unit recharging cost vary. Partial recharging policy aims to satisfy time-dependent constraints and it is in harmony with practice to not incurring a needless cost. Our second contribution is to develop a mixed integer VRP model with mixed fleet composed of EVs and ICVs. We define new decision variables removing the need of CS duplications and guaranteeing unique trace of every single visit to CSs inspired by [Koç and Karaoğlu \(2016\)](#). This is different from the formulation approach used in the literature where the network is augmented by copying the CS nodes to represent the visits at same CS more than once (for example see [Sassi et al., 2014](#); [Goeke and Schneider, 2015](#); [Macrina et al., 2019a](#); [Macrina et al., 2019b](#)). The purpose is to solve larger-size problems optimally by reducing the complexity of the mathematical model. When we consider the formulation which duplicates the nodes, it can be easily seen that it duplicates single depot

Table 1
Benchmarking of Mixed Fleet Problems.

Reference	Algorithm	Math Model		Time Windows	Partial Recharge	Charging Technology Selection	Linear Charging	Energy Consumption Model	Polluting Emission Model
		With Dummy Nodes	Without Dummy Nodes						
Gonçalves et al. (2011)	–	●							
Sassi et al. (2014)	Heuristic	●		●	●		●		
Goeke and Schneider (2015)	Heuristic	●		●			●	●	
Macrina et al. (2019a)	Heuristic	●		●	●		●		●
Macrina et al. (2019b)	Heuristic	●		●	●		●	●	
Vincent et al. (2021)	Heuristic	●		●	●		●	●	●
This Paper	Heuristic		●	●	●	●	●	●	●

$[2*(n^E + n^C)]$ times to represent all starting and ending depots for each vehicle (n^E and n^C are the number of EVs and ICVs, respectively), and also increases the number of CS nodes in the network from $s(R)$ to $s(R') (= s(R)*(\sigma + 1))$ where $s(R)$, $s(R')$ and σ stand for the number of CSs in the original problem, the number of CS nodes after duplications and permitted number of CS copies, respectively. Thus, the complexity of our formulation decreases since it prevents these kinds of duplications. Our third contribution is to develop a tailor-based powerful ALNS algorithm with several successful integrated mechanisms. These are new insertion based constructive algorithm for initialization, neighborhood mechanisms, approaches for boosting diversification effect, scoring policy for neighborhood mechanisms, and a new enhancement procedure.

The remainder of this paper is organized as follows. Section 2 defines the problem and introduces the mathematical formulation. Section 3 describes the initialization phase and ALNS metaheuristic. Section 4 presents the computational experiments. Finally, Section 5 presents our conclusions.

2. Formal problem definition and formulation

This section introduces the problem definition by considering all parameters, functions, notations, attributes and assumptions and the mathematical model.

2.1. Problem definition and notation

The MF-VRP-MC can be formulated as follows. Let $G = (V_0 \cup R, A)$ be a complete directed graph where $V_0 = V \cup 0$ is a set of nodes in which $V = \{1, \dots, N\}$ denotes customer nodes and node 0 represents the depot, respectively. The depot is both starting and ending node for all routes. Furthermore, the set R stands for CSs to supply the recharging demand of EVs. $A = \{(i, j) | i, j \in V_0 \cup R, i \neq j\}$ is the set of arcs, each of which has a nonnegative cost proportional to predetermined distances d_{ij} and unit travelling costs related to ICVs (c_{ij}^C) or EVs (c_{ij}^E). The superscripts C and E represent the relation of parameters with ICVs and EVs, respectively. Predefined travelling time is denoted by $t_{ij} = d_{ij}/s$ where s is the average speed on arcs. The distance from customer node i to customer node j through station k is stated as $d_{ijk} = d_{ik} + d_{kj}$. In a similar way, travelling time from customer node i to customer node j through station k is denoted by $t_{ijk} = t_{ik} + t_{kj}$.

Each customer $i \in V$ has a delivery demand q_i , a service time S_i , and a time window $[e_i, l_i]$ where e_i and l_i denote the ready date and due date for service, respectively. Vehicles have to reach the customer nodes for service earlier than the due dates. However, in case of arriving the customer node earlier than the ready date, then the vehicle has to wait until the ready date. Each customer $i \in V$ has to be visited exactly once by a single vehicle either EV or ICV.

Mixed fleet is composed of limited numbers of homogeneous EVs and ICVs, which are symbolized by n^E and n^C , respectively. EVs are characterized by load capacity C^E , whereas ICVs are characterized by load capacity C^C . EVs are also defined by their battery capacity Q^E . Each vehicle serves on a single route within the planning horizon instead of considering multiple trips. Moreover, each vehicle assigned for service has to complete its tour before an imposed maximum tour duration time T_{max} .

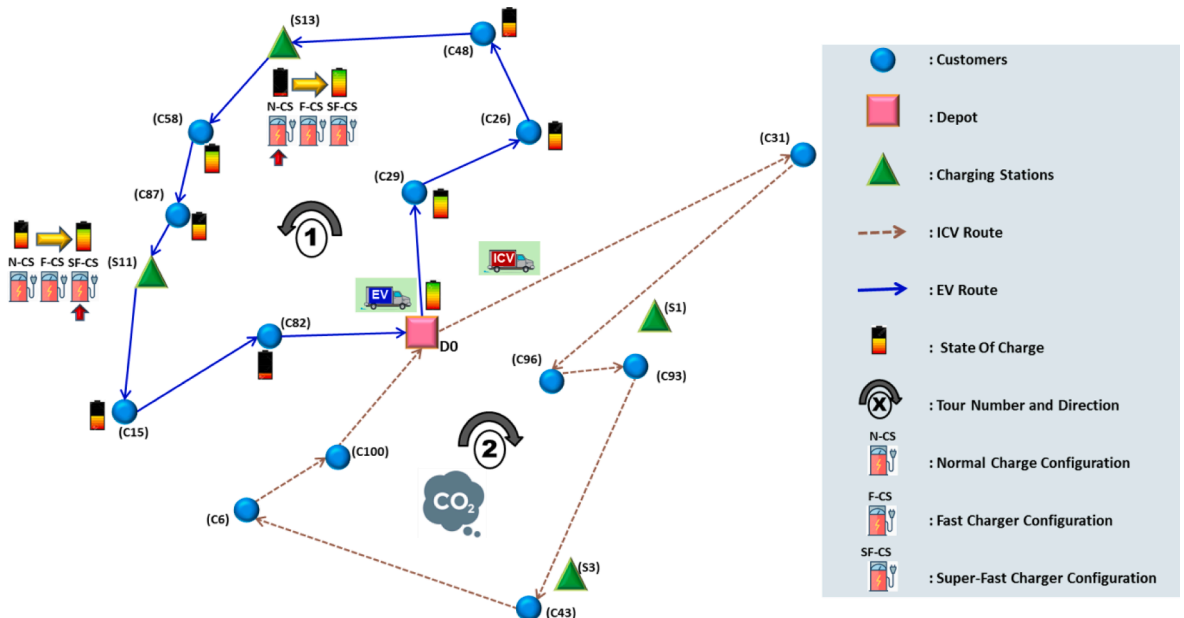


Fig. 1. An illustrative example of the MF-VRP-MC.

EVs have a limited driving range due to their battery capacity. Thus, EVs are allowed to stop by CSs at several predefined locations. CS may be visited by the same or different vehicles more than once relying on routing requirements. CSs have $m \in M$ different types of battery charger configuration whose unit recharging time g^m and in parallel unit recharging cost c^m vary. Preferring a faster charger configuration to recharge causes higher costs. Therefore, the trade-off between minimizing the recharging costs and satisfying the time related constraints is one of the challenging issues.

EVs are fully recharged at the depot and partially recharged up to the needed level at any available CS en-route. Battery charge level is assumed as a linear function of charging time. Battery charge is consumed only en-route not during service, depending on the load and travelled distance. The average quantity of charge consumption for unit distance depending on the load is a piecewise linear function which is referred to as set α consisting of load breaking points o_l as elements and the value of charge consumption rate $h(u_{ij}^E)$ which varies according to the load. The number of elements of set α is Ω .

The overall amount of CO₂ emitted by the assigned ICVs is bounded with a limitation UB . This upper bound, refers to adaptation to green logistics applications intrinsically, is expected to play a critical role on utilization rate of ICVs. Similar to energy consumption of EVs, emission function of ICVs is also based on load and travelled distance. Hence the average amount of CO₂ emission for unit distance $\varepsilon(u_{ij}^C)$ as a piecewise linear function closely relates with the set β of load breaking points o'_l , whose number of elements is Ω' .

An illustrative example of the MF-VRP-MC is given in Fig. 1. In this example, there are 1 depot, 13 customers and 4 CSs. The fleet contains one ICV and one EV in order to supply the customer demands. Thus, the EV is assigned to route 1 to serve 7 customers and the ICV is assigned to route 2 to serve 6 customers. SoC at the arrival at each node on route 1 is depicted in the figure. The EV leaves the depot as fully charged. But after the third customer (C48) on route, the EV visits the CS (S13) and recharges the battery up to full level by the normal recharge configuration. Then the EV visits another CS (S11) one more time for recharging after the fifth customer (C87). This time, super-fast recharge configuration is utilized in order to reach the next customers on route within the predefined time windows. As for the ICV, it is assigned to route 2 by considering that the total amount of CO₂ emission will not exceed UB . As there is no need to recharge, the ICV visits only the customers on route.

Decision variables of the problem are as follows.

Continuous decision variables

T_i : Service starting time at customer node $i \in V$

u_{ij}^E : Amount of load carried to node j from node i by EV, $\forall i, j \in V_0, i \neq j$

u_{ij}^C : Amount of load carried to node j from node i by ICV, $\forall i, j \in V_0, i \neq j$

y_i : Battery state of charge (SoC) at the arrival at node i , $\forall i \in V_0 \cup R$

y_i^{bd} : Battery state of charge (SoC) at the arrival at node i which is just before the ending depot on route, $\forall i \in V$ (defined for determining the value of decision variable y_i^{SR} below)

y_i^{SR} : Summation of all remaining battery state of charges belonging to assigned EVs at the arrival at depot

θ_{ijk}^m : Amount of energy recharged at station k using charger configuration m between node i and node j , $\forall i, j \in V_0, \forall k \in R, \forall m \in M, i \neq j$

Binary variables

x_{ij}^E : 1 if an EV travels on arc $(i, j) \in A$, 0 otherwise, $i, j \in V_0, i \neq j$

x_{ij}^C : 1 if an ICV travels on arc $(i, j) \in A$, 0 otherwise, $i, j \in V_0, i \neq j$

z_{ijk} : 1 if an EV travels from node i to node j through station k , 0 otherwise, $\forall i, j \in V_0, \forall k \in R, i \neq j$

a_{ijk}^m : 1 if an EV is recharged at station k between node i and node j by charger

configuration m , 0 otherwise, $\forall i, j \in V_0, \forall k \in R, \forall m \in M, i \neq j$

Auxiliary variables

p_{ijl} : 1 if amount of load on EV is within the range $[o_l, o_{l+1})$ on arc $i-j$, 0 otherwise,

$\forall i, j \in V_0, l \in \alpha \setminus \{o_\Omega\}, i \neq j$

f_{ijl} : a nonnegative continuous auxiliary variable that determines the value of unit energy consumption due to load within the range $[o_l, o_{l+1})$ on arc $i-j$, $\forall i, j \in V_0, l \in \alpha, i \neq j$

p_{ijl}^k : 1 if amount of load on EV is within the range $[o_l, o_{l+1})$ while travelling to node j from node i through station k , 0 otherwise, $\forall i, j \in V_0, \forall k \in R, l \in \alpha \setminus \{o_\Omega\}, i \neq j$

f_{ijl}^k : a nonnegative continuous auxiliary variable that determines the value of unit energy consumption due to load within the range $[o_l, o_{l+1})$ while travelling to node j from node i through station k , $\forall i, j \in V_0 \cup R, \forall k \in R, l \in \alpha, i \neq j$

p'_{ijl} : 1 if amount of load on ICV is within the range $[o'_l, o'_{l+1})$ on arc $i-j$, 0 otherwise,

$\forall i, j \in V_0, l \in \beta \setminus \{o'_\Omega\}, i \neq j$

f'_{ijl} : a nonnegative continuous auxiliary variable that determines the value of unit

emission amount due to load within the range $[o'_l, o'_{l+1})$ on arc $i-j$,

$\forall i, j \in V_0, l \in \beta, i \neq j$

2.2. Mathematical model

The mathematical model of the MF-VRP-MC is then:

$$\text{Min} \sum_{i \in V_0} \sum_{j \in V_0} \sum_{k \in R} \sum_{m \in M} c^m \theta_{ijk}^m + c^1 (Q^E (\sum_{j \in V} x_{0j}^E + \sum_{j \in V} \sum_{k \in R} z_{0jk}) - y^{SR}) + (\sum_{i \in V_0} \sum_{j \in V_0} c_{ij}^E d_{ij} x_{ij}^E + \sum_{i \in V_0} \sum_{j \in V_0} \sum_{k \in R} (c_{ik}^E d_{ik} + c_{kj}^E d_{kj}) z_{ijk}^E) + \sum_{i \in V_0} \sum_{j \in V_0} c_{ij}^C d_{ij} x_{ij}^C \quad (1)$$

$$\sum_{j \in V_0} (x_{ij}^E + \sum_{k \in R} z_{ijk}) + \sum_{j \in V_0} x_{ij}^C = 1 \forall i \in V, i \neq j \quad (2)$$

$$\sum_{i \in V_0} x_{ij}^E + \sum_{i \in V_0} \sum_{k \in R} z_{ijk} = \sum_{i \in V_0} x_{ji}^E + \sum_{i \in V_0} \sum_{k \in R} z_{jik} \forall j \in V, i \neq j \quad (3)$$

$$\sum_{i \in V_0} x_{ij}^C = \sum_{i \in V_0} x_{ji}^C \forall j \in V, i \neq j \quad (4)$$

$$\sum_{j \in V} x_{0j}^E + \sum_{j \in V} \sum_{k \in R} z_{0jk} \leq n^e \quad (5)$$

$$\sum_{i \in V} x_{i0}^E + \sum_{i \in V} \sum_{k \in R} z_{i0k} \leq n^e \quad (6)$$

$$\sum_{j \in V} x_{0j}^C \leq n^C \quad (7)$$

$$\sum_{i \in V} x_{i0}^C \leq n^C \quad (8)$$

$$\sum_{j \in V} x_{0j}^E + \sum_{j \in V} \sum_{k \in R} z_{0jk} = \sum_{i \in V} x_{i0}^E + \sum_{i \in V} \sum_{k \in R} z_{i0k} \quad (9)$$

$$\sum_{j \in V} x_{0j}^C = \sum_{i \in V} x_{i0}^C \quad (10)$$

$$T_i + (t_{ij} + S_i)x_{ij}^E + \sum_{k \in R} ((t_{ijk} + S_i)z_{ijk} + \sum_{m \in M} g^m \theta_{ijk}^m) - M(1 - x_{ij}^E - \sum_{k \in R} z_{ijk}) \leq T_j \forall i \in V_0, \forall j \in V, i \neq j \quad (11)$$

$$T_i + (t_{ij} + S_i)x_{ij}^C - M(1 - x_{ij}^C) \leq T_j \forall i \in V_0, \forall j \in V, i \neq j \quad (12)$$

$$T_i + (t_{i0} + S_i)x_{i0}^E + \sum_{k \in R} ((t_{i0k} + S_i)z_{i0k} + \sum_{m \in M} g^m \theta_{i0k}^m) \leq T_{max} \forall i \in V \quad (13)$$

$$T_i + (t_{i0} + S_i)x_{i0}^C \leq T_{max} \forall i \in V \quad (14)$$

$$e_j \leq T_j \leq l_j \forall j \in V_0 \quad (15)$$

$$\sum_{i \in V_0} u_{ij}^E - \sum_{i \in V_0} u_{ji}^E = q_j (\sum_{i \in V_0} x_{ij}^E + \sum_{i \in V_0} \sum_{k \in R} z_{ijk}) \forall j \in V, i \neq j \quad (16)$$

$$\sum_{i \in V_0} u_{ij}^C - \sum_{i \in V_0} u_{ji}^C = q_j \sum_{i \in V_0} x_{ij}^C \forall j \in V, i \neq j \quad (17)$$

$$0 \leq u_{ij}^E \leq C^E (x_{ij}^E + \sum_{k \in R} z_{ijk}) \forall i \in V_0, \forall j \in V_0, i \neq j \quad (18)$$

$$0 \leq u_{ij}^C \leq C^C x_{ij}^C \forall i \in V_0, \forall j \in V_0, i \neq j \quad (19)$$

$$0 \leq y_i \leq y_i - [\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl} h(o_l) \} d_{ij} + \sum_{k \in R} (\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl}^k h(o_l) \} d_{ijk} - \sum_{m \in M} \theta_{ijk}^m)] + Q^E (1 - x_{ij}^E - \sum_{k \in R} z_{ijk}) \forall i \in V_0, \forall j \in V, i \neq j \quad (20)$$

$$0 \leq y_i - (\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{i0l} h(o_l) \} d_{i0} + \sum_{k \in R} (\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{i0l}^k h(o_l) \} d_{i0k} - \sum_{m \in M} \theta_{i0k}^m)) \forall i \in V \quad (21)$$

$$0 \leq y_i - (\sum_{j \in V_0} \sum_{k \in R} \{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl}^k h(o_l) \} d_{ik}) \forall i \in V_0, i \neq j \quad (22)$$

$$y_i^{bd} \leq (x_{i0}^E + \sum_{k \in R} z_{i0k}) Q^E \forall i \in V \quad (23)$$

$$y_i^{bd} \leq y_i \forall i \in V \quad (24)$$

$$y^{SR} = \sum_{i \in V} (y_i^{hd} - [\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{i0l} h(o_l) \} d_{i0} + \sum_{k \in R} (\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{i0l}^k h(o_l) \} d_{i0k} - \sum_{m \in M} \theta_{i0k}^m)]) \quad (25)$$

$$y_i - \sum_{j \in V_0} \sum_{k \in R} (\{ \sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl}^k h(o_l) \} (d_{ik}) - \sum_{m \in M} \theta_{ijk}^m) \leq Q^E \forall i \in V_0, i \neq j \quad (26)$$

$$y_0 = Q^E \quad (27)$$

$$u_{ij}^E = \sum_{l \in \alpha} f_{ijl}(o_l) + \sum_{k \in R} \sum_{l \in \alpha} f_{ijl}^k(o_l) \forall i \in V_0, \forall j \in V_0, i \neq j \quad (28)$$

$$f_{ij1} \leq p_{ij1} \forall i \in V_0, \forall j \in V_0, i \neq j \quad (29)$$

$$f_{ijl} \leq p_{ijl} + p_{ijl-1} \forall i \in V_0, \forall j \in V_0, \forall l \in \{2 \dots \Omega - 1\}, i \neq j \quad (30)$$

$$f_{ijl} \leq p_{ijl-1} \forall i \in V_0, \forall j \in V_0, l = \Omega, i \neq j \quad (31)$$

$$\sum_{l \in \alpha} f_{ijl} = x_{ij}^E \forall i \in V_0, \forall j \in V_0, i \neq j \quad (32)$$

$$\sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl} = x_{ij}^E \forall i \in V_0, \forall j \in V_0, i \neq j \quad (33)$$

$$f_{ij1}^k \leq p_{ij1}^k \forall i \in V_0, \forall j \in V_0, \forall k \in R, i \neq j, \quad (34)$$

$$f_{ijl}^k \leq p_{ijl}^k + p_{ijl-1}^k \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall l \in \{2 \dots \Omega - 1\}, i \neq j \quad (35)$$

$$f_{ijl}^k \leq p_{ijl-1}^k \forall i \in V_0, \forall j \in V_0, \forall k \in R, l = \Omega, i \neq j \quad (36)$$

$$\sum_{l \in \alpha} f_{ijl}^k = z_{ijk} \forall i \in V_0, \forall j \in V_0, \forall k \in R, i \neq j \quad (37)$$

$$\sum_{l \in \alpha \setminus \{o_\Omega\}} p_{ijl}^k = z_{ijk} \forall i \in V_0, \forall j \in V_0, \forall k \in R, i \neq j \quad (38)$$

$$0 \leq \theta_{ijk}^m \leq Q^E a_{ijk}^m \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall m \in M, i \neq j \quad (39)$$

$$\sum_{m \in M} a_{ijk}^m = z_{ijk} \forall i \in V_0, \forall j \in V_0, \forall k \in R, i \neq j \quad (40)$$

$$\sum_{i \in V_0} \sum_{j \in V_0} (\sum_{l \in \beta \setminus \{o'_\Omega\}} p'_{ijl} \mathcal{E}(o'_l)) d_{ij} \leq UB \quad (41)$$

$$u_{ij}^C = \sum_{l \in \beta} f'_{ijl}(o'_l) \forall i \in V_0, \forall j \in V_0 \quad (42)$$

$$f'_{ij1} \leq p'_{ij1} \forall i \in V_0, \forall j \in V_0, i \neq j \quad (43)$$

$$f'_{ijl} \leq p'_{ijl} + p'_{ijl-1} \forall i \in V_0, \forall j \in V_0, \forall l \in \{2 \dots \Omega' - 1\}, i \neq j \quad (44)$$

$$f'_{ijl} \leq p'_{ijl-1} \forall i \in V_0, \forall j \in V_0, \forall l = \Omega', i \neq j \quad (45)$$

$$\sum_{l \in \beta} f'_{ijl} = x_{ij}^C \forall i \in V_0, \forall j \in V_0, i \neq j \quad (46)$$

$$\sum_{l \in \beta \setminus \{o'_\Omega\}} p'_{ijl} = x_{ij}^C \forall i \in V_0, \forall j \in V_0, i \neq j \quad (47)$$

$$x_{ij}^E, z_{ijk} \in \{0, 1\} \forall i \in V_0, \forall j \in V_0, \forall k \in R \quad (48)$$

$$x_{ij}^C \in \{0, 1\} \forall i \in V_0, \forall j \in V_0 \quad (49)$$

$$p_{ijl}, p_{ijl}^k \in \{0, 1\} \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall l \in \alpha \setminus \{0_\Omega\} \quad (50)$$

$$p'_{ijl} \in \{0, 1\} \forall i \in V_0, \forall j \in V_0, \forall l \in \beta \setminus \{0'_\Omega\} \quad (51)$$

$$f_{ijl}, f_{ijl}^k \geq 0 \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall l \in \alpha \quad (52)$$

$$f'_{ijl} \geq 0 \forall i \in V_0, \forall j \in V_0 \forall l \in \beta \quad (53)$$

$$u_{ij}^E, u_{ij}^C, T_i, y_i, \theta_{ijk}^m \geq 0 \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall m \in M. \quad (54)$$

$$a_{ijk}^m \in \{0, 1\} \forall i \in V_0, \forall j \in V_0, \forall k \in R, \forall m \in M. \quad (55)$$

$$y_i^{bd}, y_i^{SR} \geq 0 \forall i \in V \quad (56)$$

The objective function (1) aims to minimize the total cost consisting of five terms. The first term computes the total charging cost. The second term stands for the fully recharging cost by slow chargers at the depot for used EVs. Third component is associated with the end SoC which remains after the completion of tour by EVs. It is deducted from the cost caused by recharging at the depot. Fourth and fifth terms compute the total routing costs of EVs and ICVs, respectively.

Constraints (2) guarantee that each customer is visited exactly once by either EV or ICV. Constraints (3–4) ensure flow conservations on customers for EVs and ICVs, respectively. Constraints (5–8) prevent the number of activated EVs and ICVs from exceeding the fleet size, respectively. Constraints (9–10) ensure that each EV and ICV departing from the depot returns to the depot, respectively. Constraints (11–12) deal with service start times of vehicles. Constraints (13–14) provide all EV and ICV routes to be completed within the maximum tour duration, respectively. Constraints (15) guarantee that service starting times at customers to be between the predefined bounds of time windows. Constraints (16–17) deal with the amount of loads carried on arcs whereas this amount of load is limited up to the vehicle capacity by constraints (18–19). Constraints (20) keep track of the SoC on route whereas having sufficient SoC to reach to the depot is ensured by constraints (21). Constraints (22) enforce that SoC is enough to reach a CS in case of getting the next customer through that CS. Constraints (23–25) compute the remaining SoCs of EVs at the end of the route. Constraints (26) ensure that post-charging SoC does not exceed the battery capacity when recharging is performed over a SoC at the arrival of CS. Constraint (27) state that each EV departs the depot as fully charged.

Constraints (28) are the representations of amount of EV load on an arc in a linear affine combination form of load breaking points which compute the unit energy consumption rate. Constraints (29–33) contribute to set unit energy consumption rate with respect to load directly carried between customers. Constraints (34–38) compute unit energy consumption rate with respect to load carried between customers with a stop at CS. Constraints (39–40) impose that only one type of charger configuration is used for charging. Constraints (41) restrict the total amount of CO₂ emission. Constraints (42) represent the amount of ICV load on an arc in a linear affine combination form of load breaking points which alter the unit emission value. Constraints (43–47) enforce the assignment of unit emission value.

Finally, constraints (48–56) define the domains of decision variables.

3. Adaptive large neighborhood search

To efficiently solve the introduced problem, we develop an ALNS based metaheuristic. Many versions of the ALNS have been successfully applied to rich variants of the VRP (see Ropke and Pisinger, 2006a; Ropke and Pisinger, 2006b; Keskin and Çatay, 2018; Koç et al., 2019; Keskin et al., 2021; Turkeş et al., 2021). Our algorithm is partitioned into two phases, initialization and ALNS.

3.1. Initialization phase

We introduce an initialization phase. This phase firstly applies a clustering procedure on the customers $i \in V$ to be labeled with i_C and i_E , which refer the customers to be served on ICV and EV route, respectively. Two discrete sets V_C ($\forall i_C \in V_C$) and V_E ($\forall i_E \in V_E$) are then obtained in order to carry on two-staged insertion heuristic on these sets in a sequence. Two-stage insertion heuristic constructs the initial solution which is later improved by the ALNS. The flow chart of initialization phase is given in [supplementary material](#).

3.1.1. Clustering procedure

Within the frame of clustering procedure, two scores p_i^E and p_i^C are calculated. These scores are weighted average of several sub-scores which are based on distance, quantity of demand and ready date. These label the unlabeled customers $i \in V^u$ with labels i_C or i_E due to these scores. Set of unlabeled customers V^u is equivalent to V at the beginning of the procedure. The score p_i^E is calculated as follows.

$$p_i^E = w_1(pDist_i^E) + w_2(pQ_i) + w_3(pStaDist_i^E) + w_4(pRD_i) \quad (57)$$

The terms referring the sub-scores weighted with different coefficients (w_1, w_2, w_3, w_4) in equation (57) are given below ($\sum_{j \in \{1,2,3,4\}} w_j = 1$). It should be noted that sub-score values vary from 1 to 10.

$$pDist_i^E = 11 - (1 + (d_i^E - d_{min}^E)/(d_{max}^E - d_{min}^E)*9) \quad (58)$$

where d_i^E is the distance between the customer $i \in V^u$ and the barycenter b_E (centroid which is arithmetic mean position of all labelled customers $i_E \in V_E$). d_{min}^E denotes the distance between b_E and the customer $j^+ \in V^u$ nearest to b_E . d_{max}^E denotes the distance between b_E and the customer $j^- \in V^u$ farthest to b_E .

$$pQ_i = 11 - (1 + (q_i - q_{min})/(q_{max} - q_{min})*9) \quad (59)$$

where q_i refers the amount of demand belongs to unlabeled customer $i \in V^u$. q_{min} and q_{max} represent minimum and maximum amount of demands of all unlabeled customers $j \in V^u$, respectively.

$$pStaDist_i^E = 11 - (1 + (Sd_i^E - Sd_{min})/(Sd_{max} - Sd_{min})*9) \quad (60)$$

where Sd_i^E is the distance between the unlabeled customer $i \in V^u$ and the station $k^* \in R$ closest to the lastly added customer j_E^{last} to the set V_E . Sd_{min} denotes the distance between the station k^* and the unlabeled customer $i^+ \in V^u$ which is the nearest to station k^* . Sd_{max} denotes the distance between station k^* and the unlabeled customer $i^- \in V^u$ which is the farthest to station k^* .

$$pRD_i^E = 11 - (1 + (RD_i^E - RD_{min}^E)/(RD_{max}^E - RD_{min}^E)*9) \quad (61)$$

where RD_i^E symbolizes the time interval from the ready date of unlabeled customer $i \in V^u$ to the average ready date of all labelled customers assigned to V_E before, $\overline{RD}_{j_E \in V_E}^E \cdot RD_{min}^E$ denotes for the time interval from $\overline{RD}_{j_E \in V_E}^E$ to the unlabeled customer's $i^+ \in V^u$ ready date which is the closest to that average. RD_{max}^E denotes the time interval from $\overline{RD}_{j_E \in V_E}^E$ to the unlabeled customer's $i^- \in V^u$ ready date which is the most distant to that average.

We introduce sub-scores (59–61) over work by [Macrina et al. \(2019a\)](#). Sub-score (59) reflects the load effect in clustering procedure. Because more load may mean more energy consumption. Sub-score (60) is integrated into the total score with the motivation of that being relatively close to a CS will be determiner for customers to be on EV route. Finally, sub-score (61) is motivated for integration to the total score by that in some instances ready date of customers may majorly influence the pattern of the route besides the distances between customers.

The score p_i^C is calculated as follows.

$$p_i^C = w_5(pDist_i^C) + w_6(pQ_i) + w_7(pRD_i) \quad (62)$$

Furthermore, the terms standing for the sub-scores weighted with different coefficients (w_5, w_6 and w_7) in equation (62) are stated below ($\sum_{j \in \{5,6,7\}} w_j = 1$). Also, sub-score values vary from 1 to 10.

$$pDist_i^C = 11 - (1 + (d_i^C - d_{min}^C)/(d_{max}^C - d_{min}^C)*9) \quad (63)$$

$$pRD_i^C = 11 - (1 + (RD_i^C - RD_{min}^C)/(RD_{max}^C - RD_{min}^C)*9) \quad (64)$$

Sub-score (63) and introduced sub-score (64) are calculated similarly to (58) and (59), respectively.

In beginning of the clustering procedure, depot is first assigned to both sets V_E and V_C . At each iteration, the customer with the highest score $i^* = \text{argmax}_{i \in V^u} \{p_i^E\}$ and the customer with the highest score $j^* = \text{argmax}_{j \in V^u} \{p_j^C\}$ are selected to be labelled by i_E or i_C , respectively. If $i^* \neq j^*$, i^* is then removed from V^u and assigned to V_E where j^* is removed from V^u and assigned to V_C . If $i^* = j^*$, i^* is then assigned to either V_E or V_C according to the cases $p_i^E > p_j^C$ or $p_i^E \leq p_j^C$, respectively.

The iterations are repeated until $V^u = \emptyset$. At the beginning of each iteration the barycenter (b_E and b_C) and the average ready dates ($\overline{RD}_{j_E \in V_E}^E$ and $\overline{RD}_{j_C \in V_C}^C$) of each cluster (V_E and V_C) are recomputed. The CS $k^* \in R$ closest to the lastly added customer j_E^{last} to the set V_E is redetermined. At the end of the clustering procedure, V_E and V_C are returned.

3.1.2. Insertion procedure

In the two-staged insertion procedure, the former stage is applied on the set V_C , the latter is then applied on the set V_E . This may include customers $i_C \in V_C$ who are not feasibly routed owing to the enforcement of ICV constraints and are transferred to the set V_E . Customers in set V_E who are not feasibly routed in the second stage are transferred to set V_C . These unrouted customers are evaluated by the preassigned ICVs or a new one with the possibility of several constraint violations in case of necessity.

3.1.2.1. First stage (ICV Routes). The insertion procedure initializes the route by considering to locate the unrouted customer $k_C^* \in V_C$ between the start and end depot considering the smallest evaluation criterion value of all that are sorted in ascending order. When the feasible solution is obtained, the ICV route becomes $R_{cr}^C = \{0, k_C^*, 0\}$, where the set R_{cr}^C denotes the constructed ICV route consisting of ordered customers and cr is the index of constructed ICV routes. The routed customer k_C^* is then removed from the set V_C ($V_C \leftarrow V_C \setminus \{k_C^*\}$, $R_{cr}^C \leftarrow R_{cr}^C \cup \{k_C^*\}$).

$$k_C^* = \text{argmin}_{k \in V_C | i, j \in R_{cr}^C, |A_{ij}|=1} \{d_k = d_{ik} + d_{kj} - d_{ij}\} \quad (65)$$

($A_{ij} = 1$: nodes i and j are adjacent in the cr^{th} constructed ICV route).

$$k_C^* = \arg_{1 \leq k \leq AI} \min_{k \in V_C} \{d_k = d_{ik} + d_{kj} - d_{ij}\} \quad (66)$$

Evaluation criteria determine the customer and its position on route with the least value increase. Criterion (65) considers only the distance increase. $d_k = d_{ik} + d_{kj} - d_{ij}$ is the distance increase when $k \in V_C$ is inserted between the adjacent nodes $i, j \in R_{cr}^C$. d_{min} is the minimum distance increase after computations of all $k \in V_C$ insertions in all available positions in R_{cr}^C . d_{max} is the maximum distance increase of all alternatives.

Criterion (66) randomly selects one of the first certain number λ of values by using uniform distribution from the ascendingly sorted list in a greedy randomized approach. Randomization effect may be set by the parameter λ taking value in the range $[1, AI]$, where.

$AI = s(V_C) * (s(R_{cr}^C) - 1)$ is the number of alternative insertions. $s(V_C)$ denotes the number of elements in V_C , namely the number of uninserted customers with label i_C , whereas $s(R_{cr}^C)$ represents the number of elements in R_{cr}^C . In parallel, the number of alternative positions for insertion corresponds to $(s(R_{cr}^C) - 1)$.

If time window constraints are satisfied, successive constraint checks are performed. Otherwise, following alternative insertion based on the ascendingly sorted d_k list is selected as a candidate. When all alternative insertions violate at least one of the time window constraints, then a new ICV route is evaluated. If there is no ICV in fleet, the unserved customers are transferred to set V_E as aforementioned and removed from set V_C .

Insertions are repeated by locating $k_C^* \in V_C$ between adjacent nodes i and j which are in the currently constructed ICV route R_{cr}^C . In the case of violation, procedure stops insertions and assesses new route initialization. When cr is equal to n^C at the time of constraint violation, the first stage ends and the second begins with the transfer of unserved customers $i_C \in V_C$ to set V_E .

Second stage (EV Routes).

In the context of second stage, customer insertion procedure remains the same with the approach in the first stage in keeping with the following evaluation criteria.

$$k_E^* = \argmin_{k \in V_E} \min_{i, j \in R_{cr}^E} \{d_k = d_{ik} + d_{kj} - d_{ij}\} \quad (67)$$

($A_{ij} = 1$: nodes i and j are adjacent in the cr^{th} constructed EV route).

$$k_E^* = \arg_{1 \leq k \leq AI} \min_{k \in V_E} \{d_k = d_{ik} + d_{kj} - d_{ij}\} \quad (68)$$

If the route is energy-feasible, customer insertions continue based on the number of inactivated EVs. Otherwise, CS $k_s^* \in R$ closest to the route is inserted in the EV route (69). CS insertions are repeated as long as the energy-feasibility is satisfied. Primarily, fully recharging by slow technology is performed at the inserted CS.

$$k_s^* = \argmin_{k \in R} \min_{i, j \in R_{cr}^E} \{d_k = d_{ik} + d_{kj} - d_{ij}\} \quad (69)$$

$$y_i - h(u_{ik}^E) * d_{ik} \geq 0 \quad (70)$$

$$Q^E - h(u_{kj}^E) * d_{kj} \geq 0 \quad (71)$$

$$Q^E - h(u_{kj}^E) * d_{kj} \geq y_j^{WR} \quad (72)$$

When we insert CS $k_s^* \in R$ between adjacent nodes $i, j \in R_{cr}^E$, CS has to be reachable from node i subject to SoC (70). EV is able to reach the node j after fully charge at CS $k_s^* \in R$ (71). A meaningful and serviceable insertion that causes higher SoC at node j after fully recharging than after directly reached from node i , y_j^{WR} is ensured by (72).

After attaining an energy-feasible route by CS insertion, the route is evaluated against whether time dependent constraints are still satisfied or not, depending on the recharging process. All the unserved customers by the lack of EV in fleet are inserted to the last ICV route by the same proposed insertion strategies. At the end of the second stage, unconsumed charge level is deducted.

3.2. ALNS phase

In order to improve the quality of initial solution, we use the ALNS algorithm. We inspired from several successful ALNS mechanisms in the literature (Turkes et al., 2019). Algorithm 1 presents the detailed pseudocode of the ALNS framework.

Algorithm 1: General framework of the ALNS

1:Generate a feasible initial solution by initialization phase, current solution $\leftarrow$ initial solution, best solution $\leftarrow$ initial solution

2:Assign the initial weights $w_{nm_i}^0$, initial total scores $\pi_{nm_i}^0$ and initial number of calls $\theta_{nm_i}^0$ to the

neighborhood mechanisms ($nm_i \in NM$), set the segment length s , set the number of iterations to reset weights $n_{annihilate}$, set the number of iterations to reinitialize $n_{reinitialize}$, set the number of iterations to reheat $n_{NoImp.Heat}$, set the initial temperature T_{init} , set $n_{Ter.}$, set $n_{NoImp.Ter.}$

3:# of Iters $\leftarrow 1$, $r_{NoImp.Ter.} \leftarrow 0$, $r_{NoImp.Heat} \leftarrow 0$, # of NoNeighbour1 $\leftarrow 0$, # of NoNeighbour2 $\leftarrow 0$,

(continued on next page)

(continued)

Algorithm 1: General framework of the ALNS

```

# of Generations  $\leftarrow 0$ ,  $N_{reheating} \leftarrow 1$ .
4: while ( $r_{NoImp.Ter.}$  is less than  $n_{NoImp.Ter.}$ ) and ( $\#$  of Iters is less than  $n_{Ter.}$ ) do
5: if (( $\#$  of Iters mod segment length  $s$ ) is equal to 0) then
6: Update the weights  $w_{nm_i}^s$ 
7: Reset total scores  $\pi_{nm_i}^s$  and number of calls  $\theta_{nm_i}^s$ 
8: end if
9: if ( $\#$  of NoNeighbour1 is equal to  $n_{annihilate}$ ) then
10: Assign the initial weights  $w_{nm_i}^0$ , initial total scores  $\pi_{nm_i}^0$  and initial number of calls
     $\theta_{nm_i}^0$  to the neighborhood mechanisms ( $nm_i \in NM$ )
11:  $\#$  of NoNeighbour1  $\leftarrow 0$ 
12: end if
13: if ( $\#$  of NoNeighbour2 is equal to  $n_{reinitialize}$ ) then
14: current solution  $\leftarrow$  initial solution
15:  $\#$  of NoNeighbour2  $\leftarrow 0$ 
16: end if
17: Select the  $nm_i$  by roulette wheel strategy ( $w_{nm_i}^s$ )
18:  $\theta_{nm_i}^s$  (for selected  $nm_i$ )  $\leftarrow \theta_{nm_i}^s$  (for selected  $nm_i$ ) + 1
19: Apply selected  $nm_i$  on the current solution
20: if (neighbour solution is able to be generated subject to feasibility checks) then
21: Apply enhancement procedure in Algorithm 2
22: Accept or reject the neighbour as a new current solution by SA procedure ( $\Delta$ ,  $T$ )
23: if (neighbour solution is better than the best solution) then
24: increase the total score  $\pi_{nm_i}^s$  of the selected  $nm_i$  by  $S_1$ 
25: current solution  $\leftarrow$  neighbour solution
26: best solution  $\leftarrow$  current solution
27:  $r_{NoImp.Ter.} \leftarrow 0$ 
28:  $r_{NoImp.Heat.} \leftarrow 0$ 
29: end if
30: else then
31:  $r_{NoImp.Ter.} \leftarrow r_{NoImp.Ter.} + 1$ 
32:  $r_{NoImp.Heat.} \leftarrow r_{NoImp.Heat.} + 1$ 
33: end else
34: if (neighbour solution is better than the current solution but worse than the best
    solution) then
35: increase the total score  $\pi_{nm_i}^s$  of the selected  $nm_i$  by  $S_2$ 
36: current solution  $\leftarrow$  neighbour solution
37: end if
38: if (neighbour solution is worse than the current solution but accepted via SA
    procedure) then
39: increase the total score  $\pi_{nm_i}^s$  of the selected  $nm_i$  by  $S_3$ 
40: current solution  $\leftarrow$  neighbour solution
41: end if
42:  $\#$  of NoNeighbour1  $\leftarrow 0$ 
43:  $\#$  of NoNeighbour2  $\leftarrow 0$ 
44:  $\#$  of Generations  $\leftarrow \#$  of Generations + 1
45: end if
46: else then
47: keep the current solution unchanged
48: decrease the total score  $\pi_{nm_i}^s$  of the selected  $nm_i$  by  $S_4$ 
49:  $\#$  of NoNeighbour1  $\leftarrow \#$  of NoNeighbour1 + 1
50:  $\#$  of NoNeighbour2  $\leftarrow \#$  of NoNeighbour2 + 1
51: end else
52: if ( $\#$  of Generations mod  $n_{cool.}$  is equal to 0) then
53:  $T = T^*c$ 
54: end if
55: if ( $r_{NoImp.Heat.}$  mod  $n_{NoImp.Heat.}$  is equal to 0) then
56:  $T = T_{init} * ((100 - N_{reheating} * 5) / 100)$ 
57:  $r_{NoImp.Heat.} \leftarrow 0$ 
58:  $N_{reheating} \leftarrow N_{reheating} + 1$ 
59: end if
60:  $\#$  of Iters  $\leftarrow \#$  of Iters + 1
61: end while
62: return best solution

```

We utilize two approaches to generate neighborhood solutions. These approaches contain several destroy and repair neighborhood

mechanisms $nm_i \in NM$ where NM is the set of mechanisms. The first approach is based on transferring the customers from one route to another. We develop 48 neighborhood mechanisms. These are based on intra-transfers of customers from ICV routes to another ICV route, from EV routes to another EV route, and between EV or ICV routes. The second approach is based on activating a new route. We develop 8 different neighborhood mechanisms. Four of them extract the customers randomly or greedily from ICV routes to construct a new route randomly. Four of them remove customers from EV routes to construct a new route.

Single neighborhood mechanism $nm_i \in NM$ is introduced as a paired combination of one removal and one insertion operator. The detailed list of these mechanisms is given in the [supplementary material](#). Mechanism selection is performed by the roulette wheel strategy (73). The sector sizes of the roulette wheel are adjusted by the mechanism weight $w_{nm_i}^s$. This computes the probability $P(nm_i)$ of i 'th mechanism nm_i to be selected among the others.

$$P(nm_i) = w_{nm_i}^s / \sum_{i \in NM} w_{nm_i}^s \quad (73)$$

Weights are updated at the start of every segments "s" which are composed of predetermined number of consecutive iterations (74). The current weight of nm_i after preceding segment s is denoted by $w_{nm_i}^{s+1}$. Reaction factor r is between $[0,1]$ which adjusts the degree of including the last segment performance. $\pi_{nm_i}^s$ denotes the total score of nm_i at the end of the segment s . $\theta_{nm_i}^s$ is the number of times nm_i has been selected among the others until the start of the segment $s + 1$.

$$w_{nm_i}^{s+1} = (1 - r)w_{nm_i}^s + (r)\pi_{nm_i}^s / \theta_{nm_i}^s \quad (74)$$

If the nm_i is selected during an iteration, then $\pi_{nm_i}^s$ is increased by following scores and $\theta_{nm_i}^s$ is increased by one.

- S_1 , If generated neighbour solution outperforms the best solution of all.
- S_2 , If generated neighbour solution outperforms the current solution of that iteration.
- S_3 , If generated neighbour solution is not able to outperform the current solution but accepted via simulated annealing (SA) procedure.
- S_4 , If selected mechanism is not able to generate neighbour solution on account of unsatisfied feasibility conditions. (S_4 has to have a negative effect to lower the selection probability of the neighborhood mechanism)

If the generated neighbor solution outperforms the current one, it becomes the new current solution. Otherwise, if it is worse than the current solution, then it may be accepted as a current one with a certain probability $P_{accept} = e^{-\Delta/T}$ by simulated annealing. Δ is equal to $(Z^{neighbour} - Z^{current})$ which is the distance of neighbor solution quality. Initial temperature T_{init} is gradually decreased by cooling schedule $T_{n+1} = T_n * c$ at every cycle n_{cool} . In case of no improvement on the best solution for a predetermined number of iterations $n_{NoImp.Heat.}$, the temperature is set by $T = T_{init} * ((100 - N_{reheating} * 5) / 100)$. $N_{reheating}$ is the number of times the temperature has been increased. Cooling ratio c is set between 0 and 1.

In order to diversify solutions, if a neighbour cannot be generated at $n_{annihilate}$ number of iterations, the existing weights of neighborhood mechanisms set to the initial weight. In this way it is ensured that all neighborhood mechanisms get the same probability to be selected again. This assumes that outstanding mechanisms cannot work on current solution successfully anymore. Similarly, if a neighbour cannot be generated at $n_{reinitialize}$ number of iterations, the current solution is returned to the initial solution. This renders the algorithm incorporates multi-starts.

Two termination criteria are applied: i) there is no improvement on the best solution on successive $n_{NoImp.Ter.}$ iterations, ii) $n_{Ter.}$ total number of iterations reaches the predefined limit. Overall, ALNS improves the initial solution iteratively until one of the termination criteria is satisfied. The best solution is returned when the algorithm terminates.

Infeasible solutions are not allowed to be accepted as a new current solution in course of the algorithm. Because repair may not successfully work after several destroy operators for our very constrained problem. We see that allocating the most of computational effort to achieve a feasible solution is more advantageous. Our neighborhood mechanisms in hold have flexibility of generating alternatives by switching the nodes, positions and routes to reach a feasible neighbour. Those alternatives are evaluated through the constraint checks. If no feasible solution is attained on all evaluated alternatives, then infeasible solutions are disregarded. The algorithm continues with the last obtained feasible solution. Only EV routes are repaired by greedily locating CSs to maintain energy feasibility. Furthermore, if there are time window violations, charging configurations are then changed to higher technologies.

3.2.1. Enhancement procedure

Some generated solutions may contain redundant CS visits or more than needed amount of energy transfers due to considering only customer movements while generating neighbours. To successfully tackle with those redundancies, we introduce an enhancement procedure. The detailed pseudocode is provided in the [supplementary material](#).

In the procedure, modified routes are backtraced by examining the SoC on customer nodes which are just before CS. If SoC in the $(i)^{th}$ order is adequate to reach the next CS or depot without visiting the adjacent CS in the $(i + 1)^{th}$ order, then that visited CS in the $(i + 1)^{th}$ order is redundant. It is removed from the solution. If visiting the adjacent CS in the $(i + 1)^{th}$ order is mandatory to reach the next CS or depot, amount of energy transfer at the adjacent CS is then evaluated. Amount of energy transferred at the CS in the $(i + 1)^{th}$ order is set down to strict minimum needed level to reach the next CS or depot.

3.2.2. Neighborhood operators

Four removal and five insertion operators are used in the ALNS. Each of them is marked by labels. The label is constructed firstly by uppercase letter which characterizes the approach of neighbour generation mentioned in Section 3.2. The letter “C” is for “intra-transfers of customers between ICV routes”, “E” is for “intra-transfers of customers between EV routes”, “CE” is for “inter-transfers of customers between different typed routes”, and “A” is for “activating a new route”. If the label is “CE(3B)”, it means that “CE” inter-transfers of customers between different vehicles is performed by using removal operator (3) and insertion operator (B). Activating a new route is denoted by a lowercase letter where “c” is for ICV route and “e” is for EV route. The list of neighborhood operators and their pseudocodes is provided in the [supplementary material](#). Differently from the literature, greediness and randomness may be separately applied on the selection of either route or customer during both removal and insertion, rather than holistic approach.

We first define removal operators.

Random customer removal (1): This operator first determines ψ ICV or EV routes to remove customers where ψ is randomly determined. If the label is “C”, ψ ranges in $[1, R^C * RRP]$ where R^C is the number of ICV routes in current solution and RRP is the route removal percentage. If the label is “E”, ψ ranges in $[1, R^E * RRP]$, where R^E is the number of EV routes in current solution and RRP is the route removal percentage. If the label is “CE”, ψ is set to 1. Then χ customers are randomly removed from selected ψ routes and added to the removal list $L_{removals}$. χ ranges in $[1, s(R_{cr}) * CRP]$ where $s(R_{cr})$ is the number of customers in the selected route and CRP is the customer removal percentage.

Random route selection and worst-distance customer removal (2): In this operator, χ customers are removed from each of the selected ψ ICV or EV routes in a worst-distance greedy manner rather than randomly.

Tightest route selection and worst-distance customer removal (3): This operator selects the tightest route to remove the customers by means of the proposed tightness index (75). We introduce this strategy to remove customers from tighter routes and insert into looser routes ($\sum_{j \in \{8,9,10\}} w_j = 1$).

$$Tightness_R = w_8(T_0^R/T_{Max}) + w_9(s(V_{afterEDD}^R)/s(V^R)) + w_{10}(u_{0j}^R/C^R) \quad (75)$$

where $Tightness_R$ symbolizes the tightness index of route R . The first component in equation is the ratio of completion time of route R , T_0^R , to permitted maximum route duration time T_{Max} . The second component is the ratio of the number of customers at which are arrived after early due dates, $s(V_{afterEDD}^R)$, to the number of all customers in route R , $s(V^R)$. The third component is the ratio of the carried load on vehicle leaving the depot, u_{0j}^R , to the load capacity of the vehicle assigned to route R , C^R .

Worst-distance customer removal (4): This operator first calculates the number of customers χ to be removed from the solution. χ ranges in $[1, s(V) * CRP]$. $s(V)$ is the number of customers in the problem and CRP is the customer removal percentage. The customers are then removed from the solution consisting of either all ICV or all EV routes in a worst-distance greedy manner and added to the removals list $L_{removals}$.

We now define insertion operators.

Random route selection and least-distance customer insertion (A): This insertion heuristic randomly determines the ICV or EV route to insert the customers. If there is no feasible customer in $L_{removals}$, new random route for insertion is selected. Otherwise, the customer is inserted into the position which causes least distance increase. If a time related constraint is violated, then the next alternative is selected until all time related constraints are satisfied. If the insertion is performed into ICV route, emission check specific to insertion is then applied. If the insertion is performed into EV route, energy-feasibility is then evaluated right after removal and insertion realization as similar to initialization phase.

Closest route selection and least-distance customer insertion (B): This operator selects ICV or EV route whose centroid is closest to selected customer for insertion. If capacity violation occurs, the selected customer is then extracted from $L_{removals}$. Otherwise, the customer is inserted into the position which causes least distance increase. When all alternative insertions violate at least one time related constraints, then a new route which is closest to next customer in $L_{removals}$ is selected. For insertions into ICV routes, emission check specific to insertion is performed. For insertions into EV routes, all the following steps in operator (A) is valid.

Loosest route selection and least-distance customer insertion (C): This operator operates in the same way as operator (B) except route selection strategy. Loosest route which has the lowest tightness index (79) is selected for insertions.

Least-distance customer insertion (D): In this operator, least-distance customer insertion is evaluated considering all customers in $L_{removals}$ which do not violate capacity constraints, all routes in solution and all positions on those routes. Time window check is then performed specific to selected insertion. When all alternative insertions violate at least one of the time related constraints, the operator then ends without generating a solution. For insertions into ICV routes, emission check is performed at last. For insertions into EV routes, all the following steps in heuristic (A) is valid. Those steps are regularly repeated until no customer is left in $L_{removals}$.

Greedy route construction (E): While generating a new solution with an operator which transfers the customers from one route to another as explained in Section 3.2, some routes may be deactivated by removal operators. This could cause deviation from the optimal solution by providing reduction in the number of the activated vehicles. Optimal solution may include more routes than the achieved one. In this operator, removals list $L_{removals}$ is first filled by one of the four removal operators. New route is greedily constructed in a similar way to the initialization phase. In the event of not achieving a feasibility, the construction is then finalized and the algorithm returns the previous feasible solution.

4. Computational experiments

In this section, to verify the performance of the ALNS algorithm, extensive computational experiments are conducted. Our aim is to

compare the quality and the run-time of solutions achieved by both CPLEX and the ALNS, and investigate the impacts of weights and several heuristic mechanisms on solution quality. The mathematical model is coded in GAMS 24.2.2 development environment and solved by CPLEX 12.6.0.0 with its default settings. The ALNS algorithm is implemented in C#. All experiments are conducted on a PC with Intel Core i5 CPU 2.5 GHz processor and 4 GB RAM.

4.1. Data set

The E-VRP with Time Windows instances are reconstructed by Schneider et al. (2014) over the well-known Solomon (1987) benchmark instances. They integrated CSs into Solomon (1987) data at randomly determined locations. However, to achieve feasible and meaningful solutions, they also restricted the region where the CSs were located. It is guaranteed that each customer is able to be reached directly from the depot with at most two intermediate CS visits. Solomon (1987) instances are clustered on account of the geographical distribution of customer locations into three sets C (clustered distribution), R (random distribution) and RC (mixture of clustered and randomly distributed divisions). While C1, R1 and RC1 denote short scheduling horizon and tight time windows, C2, R2 and RC2 denote long scheduling horizon and loose time windows. Schneider et al. (2014) set extra parameters related to EV such as CS locations, battery capacity, energy consumption and recharging rates in harmony with the time windows of customers. Test instances are named by the labels like “rc108C5”. “rc108C5” designates that the instance has a mixture of clustered and randomly distributed divisions, “rc”, short scheduling horizon and tight time windows, “1” and finally-five customers, “C5”.

In our data, the input parameters of the modified (for small- and large-size) and generated (for medium-size) instances are selectively tuned up by preliminary experiments. We aim to generate relatively balanced mission assignments to each type of vehicles in the fleet. To adapt the instances to our problem, we consider Felipe et al. (2014) charger technologies since we deal with the multiple charger configurations whose unit recharging time g^m and in parallel unit recharging cost c^m vary. We set the unit recharging times of different configurations to 0.277 h/Kwh, 0.05 h/Kwh and 0.022 h/Kwh, and unit recharging costs to 0.160 €/Kwh, 0.176 €/Kwh and 0.192 €/Kwh for slow, medium and fast chargers, respectively.

We consider the load on EVs as one of the parameters which steers the unit amount of energy consumption for unit distance instead of constantly setting equal to 1 kw/km as in Schneider et al. (2014). We set the unit amounts of energy consumption due to the carried load, $h(u_j^E)$, to 0.6, 0.8 and 1 kw/km for low, medium and high load percentages by capacity, respectively. Low load ranges between $[0, 0.25 C^E]$, medium load ranges between $[0.25 C^E, 0.75 C^E]$ and high load ranges between $[0.75 C^E, C^E]$, respectively. This stepwise function is depicted in Fig. 2.

Our problem also asserts that amount of emission is a function of load besides the distance. To overcome this complication, we follow a similar assumption of energy consumption based on load. We assumed that the unit amount of emission due to the carried load, $\epsilon(u_j^E)$, as 0.7, 0.9 and 1 kg CO₂/km for varying load percentages by capacity as the same as in energy consumption function. Fig. 3 shows that stepwise function inspired by the data of Ubeda et al. (2014) after several capacity scenarios for a 10-tone capacity vehicle.

Emission upper bounds are tuned up as follows. 100 kg CO₂/km for 5 customers, 150 and 200 kg CO₂/km for 10 and 15 customers, 250 and 300 kg CO₂/km for 25 customers, 250, 350 and 400 kg CO₂/km for 30 customers, 400, 450 and 500 kg CO₂/km for 50 customers, and finally 800, 1000 and 1200 kg CO₂/km for 100 customers. Freight capacity of all vehicles is fixed to 200 kg. This corresponds to a homogeneous fleet where the unit travelling costs c_{ij}^C and c_{ij}^E are stabilized at 1 €/km for both vehicle types.

In consideration of total amount of demands in the instances and predetermined vehicle capacity, number of vehicles of each type in fleet are ranging from 1 to 3 for small-size instances, from 2 to 8 for medium-size instances, and from 5 to 10 for.

large-size instances. Maximum tour duration is set to the due date in time windows of the depot. The rest of the parameters such as average speed, amount of demands, customer service times, time windows and the battery capacity of EVs are assumed to equal given in Schneider et al. (2014).

For instances with 5, 10 and 15 customers each, two instances from each type of instance sets (R1, C1, RC1, R2, C2 and RC2) are utilized to conduct performance tests. Furthermore, we also use two instances of each instance set type to test large-size problems. The medium-size instances are generated by extracting several customer and station nodes from the large size instances. Within the method, 25, 30 and 50 customers are consecutively extracted from the large-size instances in an unchanged order. However, the

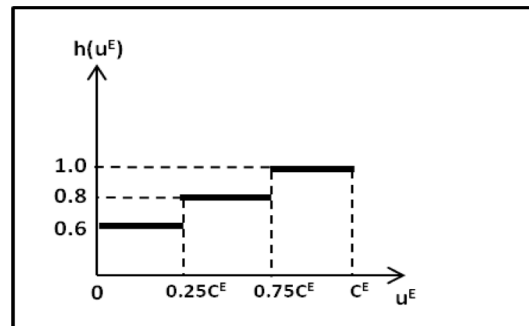


Fig. 2. Unit amount of energy consumption for unit distance.

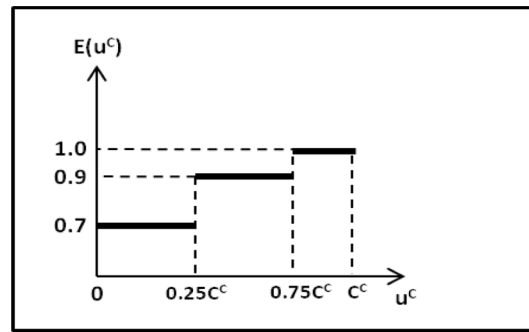


Fig. 3. Unit amount of emission for unit distance.

position where we start to extract from is randomly chosen. Suitable number of stations relatively close to the extracted customers are deliberately determined and integrated to the instances. The number of stations is set to the value in the range $[0, \lfloor N/5 \rfloor]$ where N is the number of customers. Furthermore, the depot also serves as a CS with the slow charger configuration type. In total, we generate 84 test instances composed of 36 small-, 36 medium- and 12 large-size instances.

Table 2

Results of CPLEX and ALNS on small-size instances.

Instance	CPLEX			ALNS					Δ_{ALNS}	Δ_{imp}
	Z^*	Time	Δ_{opt}	Z^{init}	Z^{ALNS}			Time		
					Min	Avg.	CV			
c101C5	253.42	0.42	0.00	253.42	253.42	253.42	0.00	0.27	0.00	0.00
c103C5	168.37	0.86	0.00	189.88	168.37	171.99	4.43	0.19	0.00	11.33
c206C5	227.19	0.89	0.00	233.03	227.19	233.03	0.00	0.07	0.00	2.51
c208C5	175.35	0.92	0.00	265.25	175.35	175.35	0.00	0.18	0.00	33.89
r104C5	142.55	1.02	0.00	178.66	142.55	143.48	0.56	0.18	0.00	20.21
r105C5	157.14	0.31	0.00	177.67	157.14	162.15	5.06	0.23	0.00	11.56
r202C5	126.52	0.55	0.00	176.28	126.52	130.28	6.29	0.22	0.00	28.23
r203C5	198.54	1.89	0.00	198.54	198.54	198.54	0.00	0.06	0.00	0.00
rc105C5	238.73	0.63	0.00	243.81	238.73	240.63	0.60	0.38	0.00	2.08
rc108C5	261.02	0.69	0.00	262.43	262.43	262.43	0.00	0.15	0.54	0.00
rc204C5	184.11	2.41	0.00	196.83	184.11	185.80	1.16	0.19	0.00	6.46
rc208C5	179.62	1.45	0.00	214.14	179.62	188.22	4.09	0.18	0.00	16.12
Average		1.00	0.00				1.85	0.19	0.04	11.03
c101C10	385.64	2.53	0.00	463.50	385.64	389.90	0.82	3.98	0.00	16.80
c104C10	259.92	48.55	0.00	306.31	259.92	259.92	0.00	17.49	0.00	15.15
c202C10	250.38	2.11	0.00	301.61	250.38	250.42	0.03	1.45	0.00	16.99
c205C10	237.24	1.72	0.00	354.58	237.24	246.56	4.56	2.45	0.00	33.09
r102C10	245.37	4.56	0.00	334.17	245.37	247.43	1.02	2.38	0.00	26.57
r103C10	196.22	901.08	0.00	226.41	196.22	196.22	0.00	1.01	0.00	13.33
r201C10	221.34	2.31	0.00	314.43	221.34	222.88	1.24	3.41	0.00	29.61
r203C10	213.65	30.27	0.00	300.85	213.65	234.42	8.18	0.36	0.00	28.98
rc102C10	397.08	2.14	0.00	457.69	397.08	408.58	2.02	2.58	0.00	13.24
rc108C10	347.15	55.73	0.00	472.74	347.15	353.55	2.69	3.63	0.00	26.57
rc201C10	320.22	3.16	0.00	417.15	320.22	347.26	4.93	1.74	0.00	23.24
rc205C10	334.45	2.63	0.00	348.66	334.45	346.90	1.28	1.20	0.00	4.08
Average		98.09	0.00				2.23	3.47	0.00	20.64
c103C15	361.20	3979.75	0.00	416.55	361.27	385.43	2.20	4.18	0.02	13.27
c106C15	270.82	4.16	0.00	581.03	270.82	275.55	1.97	11.60	0.00	53.39
c202C15	374.91	2353.36	0.00	591.70	374.91	390.43	4.00	4.48	0.00	36.64
c208C15	305.55	32.97	0.00	489.08	313.40	316.36	0.84	4.73	2.57	35.92
r102C15	387.98	20.59	0.00	496.09	387.98	390.78	0.76	9.87	0.00	21.79
r105C15	321.96	2.27	0.00	335.19	321.96	323.45	1.29	7.88	0.00	3.95
r202C15	382.23	7200.00	9.27	500.19	380.58	388.54	1.54	17.97	-0.43	23.91
r209C15	280.09	3218.92	0.00	334.13	280.09	283.00	2.74	11.27	0.00	16.18
rc103C15	406.33	7200.00	5.59	691.38	406.33	408.22	0.56	9.63	0.00	41.23
rc108C15	396.89	7200.00	35.14	425.66	385.10	389.68	1.97	2.46	-2.97	9.53
rc202C15	413.75	151.03	0.00	517.14	420.89	425.19	1.07	13.19	1.73	18.61
rc204C15	325.92	4185.94	0.00	459.57	329.14	334.06	1.23	31.03	0.99	28.38
Average		2918.18	4.17		1.68			10.69	0.16	25.23

4.2. Comparative analysis

We now present a comparative analysis of the results of the ALNS and the mathematical formulation. For solving formulation with CPLEX, a common time limit of two hours was imposed on the solution time for all instances. We present the detailed computational results of CPLEX on instances in the [supplementary material](#). Meantime, ALNS parameters which are fine tuned up after several preliminary experiments, are presented in the [supplementary material](#).

Tables 2-4 present the comparative results of CPLEX and ALNS, and also the improvement rates of initial solutions by ALNS. In the tables, the first column shows instance label. The next three columns report optimal or the best upper bound (Z^*), computation time (in seconds) and the optimality gap (Δ_{opt}) obtained by CPLEX. The optimality gap is the gap between the best lower bound (Z^{LB}) and the best upper bound (Z^*) when the formulation is solved by CPLEX within two hours, i.e. $\Delta_{opt} = 100 * ((Z^* - Z^{LB}) / Z^{LB})$. The next four columns report the cost of initial solution (Z^{init}), the cost of the best ALNS solution value among 10 independent runs (*Min*), average cost of these runs (*Avg.*) and the coefficient of variation (*CV*) of these run values. Finally, the last three columns report the best solution computation time (in seconds) over 10 runs of ALNS, percentage gap (Δ_{ALNS}) and percentage imp (Δ_{imp}). It should be noted that Δ_{ALNS} is the gap between the best ALNS solution and

optimal solution or the best upper bound obtained by CPLEX for a maximum of two-hours, $\Delta_{ALNS} = 100 * ((Z^{ALNS} - Z^*) / Z^*)$. Similarly, Δ_{imp} is the improvement obtained by the ALNS solution with respect to initial solution and it is calculated as.

$\Delta_{imp} = 100 * ((Z^{ALNS} - Z^{init}) / Z^{init})$. The symbol “-” in column (Z^*) means that CPLEX does not find any feasible solution within two hours. Correspondingly, (Δ_{opt}) and (Δ_{ALNS}) is not able to be computed. Table 5 also summarizes the average results of these comparisons. In addition to improved feasible solutions, the column “# of Imp.” also includes the number of instances which CPLEX fails to obtain feasible solution but ALNS succeeds. However, “Avg. Gap” and “Avg. Imp.” are computed only through the comparison with

Table 3

Results of CPLEX and ALNS on medium-size instances.

Instance	CPLEX		Δ_{opt}	Z^{init}	ALNS			Time	Δ_{ALNS}	Δ_{imp}
	Z^*	Time			Z^{ALNS}					
					Min	Avg.	CV			
c101C25	269.85	8.78	0.00	493.60	269.85	282.02	3.05	12.90	0.00	45.33
c103C25	507.26	7200.00	122.07	513.28	331.31	340.35	2.18	19.12	-34.69	35.45
c206C25	327.39	7200.00	19.98	518.55	316.03	326.28	2.08	25.31	-3.47	39.06
c208C25	327.48	7200.00	36.59	465.54	277.84	285.60	1.67	25.97	-15.16	40.32
r104C25	521.67	7200.00	58.96	513.29	407.42	418.26	3.03	20.67	-21.90	20.63
r105C25	535.67	156.34	0.00	723.30	539.87	542.39	0.77	7.56	0.78	25.36
r202C25	640.04	7200.00	72.32	652.43	469.03	475.43	1.05	26.12	-26.72	28.11
r203C25	487.90	7200.00	36.46	578.81	410.28	426.01	3.17	26.19	-15.91	29.12
rc105C25	558.01	4130.27	0.00	829.26	559.33	568.16	1.38	16.51	0.24	32.55
rc108C25	449.43	7200.00	27.91	747.21	447.55	466.97	2.65	11.90	-0.42	40.10
rc204C25	434.95	7200.00	31.11	472.45	372.53	389.66	3.53	23.04	-14.35	21.15
rc208C25	444.69	7200.00	42.03	494.39	349.84	367.21	2.55	33.19	-21.33	29.24
Average		5757.95	37.29				2.26	20.71	-12.74	32.20
c101C30	327.86	32.28	0.00	477.27	327.86	330.69	2.42	38.27	0.00	31.31
c104C30	364.40	7200.00	47.34	547.48	285.54	304.72	6.68	38.89	-21.64	47.85
c202C30	618.97	7200.00	100.52	675.56	360.07	393.40	5.06	11.89	-41.83	46.70
c205C30	446.41	7200.00	41.88	656.15	376.44	417.43	4.51	26.66	-15.67	42.63
r102C30	-	7200.00	-	858.00	668.16	674.68	0.83	26.16	-	22.13
r103C30	-	7200.00	-	889.49	580.05	600.03	1.30	28.44	-	34.79
r201C30	614.31	7106.81	0.00	1089.06	614.55	620.94	1.35	64.45	0.04	43.57
r203C30	-	7200.00	-	653.99	441.53	453.61	1.91	27.88	-	32.49
rc102C30	-	7200.00	-	843.24	529.59	554.85	3.23	47.43	-	37.20
rc108C30	-	7200.00	-	646.47	455.51	464.87	1.37	32.63	-	29.54
rc201C30	653.62	7200.00	21.18	914.97	571.71	586.34	1.17	20.18	-12.53	37.52
rc205C30	-	7200.00	-	777.79	481.03	494.59	2.42	42.69	-	38.15
Average		6594.92	35.15				2.69	33.80	-15.27	36.99
c103C50	-	7200.00	-	1077.94	489.59	507.67	2.04	42.85	-	54.58
c106C50	794.74	7200.00	61.34	1161.66	611.62	638.14	2.04	43.61	-23.04	47.35
c202C50	-	7200.00	-	943.77	569.60	583.70	1.95	39.91	-	39.65
c208C50	707.40	7200.00	64.61	732.15	533.33	540.23	0.87	94.24	-24.61	27.16
r102C50	-	7200.00	-	1654.34	1033.55	1050.98	1.80	48.82	-	37.53
r105C50	-	7200.00	-	1271.08	907.69	921.26	0.88	54.74	-	28.59
r202C50	-	7200.00	-	932.21	629.93	644.86	1.33	77.80	-	32.43
r209C50	-	7200.00	-	867.52	617.48	646.43	2.10	64.42	-	28.82
rc103C50	-	7200.00	-	1012.72	741.65	772.65	2.47	26.44	-	26.77
rc108C50	-	7200.00	-	1295.60	734.19	795.15	3.55	74.64	-	43.33
rc202C50	-	7200.00	-	1319.66	824.33	858.33	3.42	87.30	-	37.53
rc204C50	-	7200.00	-	996.03	659.96	674.36	1.43	48.10	-	33.74
Average		7200.00	62.98		1.99			58.57	-23.82	36.46

Table 4
Results of CPLEX and ALNS on large-size instances.

Instance	CPLEX			ALNS						
	Z^*	Time	Δ_{opt}	Z^{init}	Z^{ALNS}			Time	Δ_{ALNS}	Δ_{imp}
					Min	Avg.	CV			
C101_21	–	7200.00	–	2163.04	1090.60	1135.41	1.73	600.33	–	49.58
C102_21	–	7200.00	–	2102.78	1064.37	1091.91	1.65	355.71	–	49.38
C201_21	–	7200.00	–	1369.55	1009.01	1020.75	0.83	140.66	–	26.33
C202_21	–	7200.00	–	1299.12	1018.55	1039.79	1.61	112.51	–	21.60
r101_21	–	7200.00	–	2159.56	1623.14	1648.32	0.77	341.15	–	24.84
r102_21	–	7200.00	–	2078.82	1486.91	1513.01	1.60	339.15	–	28.47
r201_21	–	7200.00	–	1664.06	1235.73	1275.41	1.98	187.37	–	25.74
r202_21	–	7200.00	–	1611.03	1131.24	1164.01	1.73	221.71	–	32.02
rc101_21	–	7200.00	–	2463.60	1714.47	1742.27	1.18	643.43	–	30.41
rc102_21	–	7200.00	–	2281.02	1547.10	1587.05	1.89	290.43	–	32.12
rc201_21	–	7200.00	–	1881.19	1410.68	1457.99	2.05	347.26	–	25.01
rc202_21	–	7200.00	–	1748.38	1337.54	1376.51	1.92	159.93	–	23.50
Average		7200.00			1.58			311.64		30.57

feasibly obtained solutions by CPLEX and ALNS.

It is seen in Table 2 that optimal solutions are obtained for 33 of the 36 small-size instances by CPLEX. For the rest of 3 instances, feasible solutions are reached with an average gap 16.66 %. Average Δ_{ALNS} is 0.07 %, while 2 feasible solutions (for instances with 15 customers) obtained by CPLEX are improved by the ALNS with average 1.7 %. Moreover, our ALNS obtained 28 optimal solutions (out of 33) whose optimality are proven by CPLEX for small-size. The ALNS is nearly 100 times faster than CPLEX on average, while the average improvement obtained by ALNS with respect to initial solution is 18.97 %. The average coefficient of variation (CV) between the solution values of 10 independent runs by ALNS is 1.92 for small-size. It is concluded that ALNS is more advantageous than CPLEX in terms of computation time while reaching optimal solutions for most of the small-size samples.

As seen in Table 3, CPLEX optimally solves only 5 out of 36 instances for medium-size problems. The average gap for 15 feasible solutions is 52.29 %. It does not find any feasible solution for 16 instances in two hours. The ALNS improves 15 feasible solutions obtained by CPLEX (9 instances with 25 customers, 4 instances with 30 customers and 2 instances with 50 customers) with an average improvement of 19.55 %. Meantime, average improvement of the ALNS over all instances is around 15 %. The ALNS reaches optimal solutions for 2 instances. Meantime, the ALNS is nearly 200 times faster than CPLEX on average (37.69 s versus 6517,62 s), while the average improvement obtained by ALNS with respect to initial solution is 35.22 %. The CV between the solution values of 10 independent runs by ALNS is 2.31, which is very close to CV for small-size, 1.92. Consequently, these results show that ALNS is more advantageous than CPLEX on both solution quality and computation time for medium-size.

Table 4 presents computational results for large-size instances. As seen in Table 4, CPLEX does not find any feasible solution within two hours for large-size instances; on the contrary, our ALNS attains feasible solutions for all instances in reasonable time durations. The average computation of ALNS is around 312 s such that the ALNS is nearly 25 times faster than CPLEX on average and the ALNS improves initial solutions by around 30.57 %. Meanwhile, the CV between the solution values of 10 independent runs by ALNS is 1.58. When this value is compared with those obtained for small- and medium-size instances (i.e. 1.92 and 2.31, respectively), it can be easily seen that the CVs are very close to each other. It means that the solution quality of ALNS does not change depending on the size of problem. In summary, it is concluded that ALNS outperforms CPLEX in terms of solution quality and computation time.

A summary of computational results is given in Table 5. As this table shows that the ALNS solves 30 instances to optimally (i.e. 28 small-size and 2 medium-size instances). The average gap of the ALNS for 8 instances whose optimality are proven by CPLEX is around

Table 5
Summary of the comparison of CPLEX and the ALNS.

Instance Size	Solution Type (Time)	CPLEX	ALNS	# of Imp.	Average			
					Optimality Gap	Improvement on UB	Δ_{imp}	CV
Small (36)	No Feasible	–	–	2	1.17	1.7	18.97	1.92
	Feasible	3	8					
	Optimum (Time sec.)	33 (1005.76)	28 (4.78)					
Medium (36)	No Feasible	16	–	31	0.35	19.55	35.21	2.31
	Feasible	15	34					
	Optimum (Time sec.)	5 (6517.23)	2 (37.69)					
Large (12)	No Feasible	12	–	12	–	–	30.57	1.58
	Feasible	–	12					
	Optimum (Time sec.)	– (7200)	– (311.64)					

0.86 %. Meantime, the ALNS reaches feasible solutions for 28 instances for which CPLEX fails to find any solution within two hours. The ALNS improves upper bounds found by CPLEX for 45 instances (i.e. 2 for small-size, 31 for medium-size and 12 for large-size instances). While average optimality gaps of the ALNS are 1.17 % and 0.35 % for small- and medium-size instances, respectively, the ALNS improves the upper bounds of CPLEX for small- and medium-size instances by 1.7 % and 19.55 %, respectively. It is worthy to note that CPLEX does not find any feasible solution for large-size instances. Therefore, an average improvement of 30.57 % over the initial solution can be considered as an indicator of success for ALNS on large-size instances.

Meanwhile, the ALNS outperforms CPLEX in terms of computation time by reaching optimal or near optimal solutions in very short times (i.e. 4.78, 37.69 and 312 s for small-, medium- and large-size instances). These results prove that the proposed ALNS algorithm performs better than CPLEX in terms of solution quality and running time. In addition, the average improvements achieved by ALNS over the initial solution are 18.97 %, 35.21 % and 30.57 % for small, medium and large sized samples, respectively. These improvements are also indicators on the search capability of the proposed ALNS on the solution space to reach good quality solutions. Finally, average improvements of more than 18 % over initial solutions for all instance sizes and average CVs of less than 2.5 for all instance sizes demonstrate the robustness of the proposed ALNS.

We now detail the initial solutions achieved by the proposed constructive heuristics based on two insertion approaches. Feasible solutions are obtained for all small-size instances by at least one of the proposed insertion methods. Two of them are optimal. For medium-size instances, feasible solutions are obtained for all set as well as small-size by at least one of the proposed insertion methods. Furthermore, two CPLEX solutions are improved while 28 of instances are ended up with feasible solution, which CPLEX failed to find. At least one of the insertion methods is able to yield feasible solution for all instances.

Results show that there is no significant difference between greedy approach based on (65) and (67) and randomized greedy approach based on (66) and (68).

Greedy approach based on (65) and (67) is primarily used for initialization due to its power of deterministically yielding a feasible solution in the first run. Secondly, randomized greedy approach based on (66) and (68) is preferred for a few instances that greedy approach does not provide feasible solution.

4.3. Sensitivity analysis

We conduct a sensitivity analysis into two stages. While the first stage evaluates the effects of weights which are used in the clustering procedure for initialization, the second stage investigates the effects of acceptance probability function and reheating mechanism on the performance of the proposed ALNS.

4.3.1. Effects of weights

We firstly perform a sensitivity analysis for the weights w_1 , w_2 , w_3 and w_4 used in equation (57) and w_5 , w_6 and w_7 used in equation (62) on the clustering procedure of initialization procedure. This analysis is carried out into two steps with 21 instances selected from data sets, C, R, and RC. In the first step, we utilize following approach to determine the importance order of weights based on their

Table 6
Effects of weights on initial solution.

Instance	$w_1 : 0.0w_2, w_3 : 0.33w_4 : 0.34w_5 : 0.0w_6, w_7 : 0.5$	$w_1 : 0.34w_2 : 0.0w_3, w_4 : 0.33w_5, w_7 : 0.5w_6 : 0.0$	$w_1 : 0.34w_2, w_4 : 0.33w_3 : 0.0w_5 : 0.34w_6 : 0.33w_7 : 0.33$	$w_1 : 0.34w_2, w_3 : 0.33w_4 : 0.0w_5, w_6 : 0.5w_7 : 0.0$
c103C5	188.08	189.88	168.37	168.37
r104C5	169.44	178.66	160.91	142.55
rc108C5	262.43	262.43	262.43	262.43
c202C10	378.07	302.55	280.87	313.96
r102C10	323.40	310.67	284.17	334.17
rc102C10	517.80	444.38	457.69	476.01
c106C15	527.72	581.24	434.35	581.24
r102C15	499.38	493.92	510.83	507.86
rc103C15	681.95	EU-12*	EU-18*	634.46
c206C25	607.76	612.97	447.69	534.56
r104C25	760.33	616.08	535.19	517.15
rc108C25	691.62	707.96	480.20	709.30
c205C30	622.88	555.84	556.69	701.16
r102C30	856.56	857.34	902.06	EU-6*
rc108C30	995.12	674.16	741.61	817.21
c106C50	EU-31*	EU-12*	EU-1*	1364.39
r105C50	EU-5*	1210.81	1261.14	1232.31
rc108C50	1485.38	1220.83	1222.76	1274.23
C102_21	EU-10*	2089.60	2020.26	2132.00
r201_21	2083.39	1758.46	1739.41	1883.79
rc101_21	EU-9, TW-3, MTD-1*	EU-11, TW-5, MTD-1*	2283.03	EU-3, TW-6, MTD-1*
Total	14	9	4	13

* "EU-X": Obtained solution violates emission upper bound constraint at the rate of X%,

"TW-Y": Obtained solution violates Y of the time window constraints,

"MTD-Z": Obtained solution violates Z of the maximum tour duration constraints.

impact on the performance of the initialization procedure. Thus, while one of the weights is set to zero in an equation, others are assumed approximately equal values such that their summation is equal to 1. For example, in equation (57), if $w_1 = 0$, then $w_2 \cong w_3 \cong w_4$. Similarly, in equation (62), if $w_5 = 0$, then $w_6 = w_7 = 0.5$. As seen in Table 6, w_1 - w_5 (distance-based), w_2 - w_6 (load-based),

w_3 (CS-based) and w_4 - w_7 (ready date-based) in equations (57) and (62) are set to zero, separately. Then, the objective values of the obtained initial solutions are reported in Table 6 by using these coefficient values, respectively. The importance order of weights is determined according to the worsening of initial solution. Thus, firstly, the worst couple of solutions (written in bold in Table 6) at each line are determined and then bold values for each column is counted as seen in the last row of Table 6. These values (i.e. 14, 9, 4 and 13) are used to determine the importance order of weights for each equation. As a result, these orders for equations (57) and (62) are identified as $w_1 \geq w_4 > w_2, w_3$ and $w_5 \geq w_7 > w_6$, respectively, and $w_i \geq 0.1 \cdot x, x \in \mathbb{Z}^+$. Based on these orders, we can conclude that distance-based weights (w_1 and w_5) in both equations (57) and (62) are the most important weights and their values must be greater than or equal to ready date-based weights (w_4 and w_7). Also both distance-based and ready date-based weights must be greater than load-based and CS-based weights to obtain high quality initial solutions. In the second step, we select 20 combinations of weights based on their importance orders and then use these combinations to obtain initial solutions. Computational results are given in Fig. 4. Thus, we utilize the greedy initialization procedure to investigate the effects of selected weight combinations on the quality of the initial solution. Hence, each combination of weights is used until reaching an infeasible solution for one instance. If a combination of weights finds an infeasible solution, then this combination is not considered for the remaining instances. As seen in Fig. 4, the combinations numbered 3 and 7 obtain infeasible solution for the instance “rc108C5”. This is symbolized with a label “(3,7): EU-50, TW-1, MTD-1” on the left side of the dotted vertical line intercepting “rc108C5”. The label “(3,7): EU-50, TW-1, MTD-1” means that initial solutions obtained by the weight combinations numbered 3 and 7 violate emission upper bound constraint at the rate of 50 % (EU-50), one of the time window constraints (TW-1) and one of the maximum tour duration constraints (MTD-1).

As a result of analysis, only combinations numbered 1, 4, 10, 18 obtain feasible solutions for all instances with a negligible value gaps (coefficient of variation is 1.99). The list of these combinations are given as follows:

Combination 1: $w_1 = 0.3, w_2 = 0.2, w_3 = 0.2, w_4 = 0.3$ and $w_5 = 0.4, w_6 = 0.2, w_7 = 0.4$

Combination 4: $w_1 = 0.4, w_2 = 0.1, w_3 = 0.2, w_4 = 0.3$ and $w_5 = 0.4, w_6 = 0.2, w_7 = 0.4$

Combination 10: $w_1 = 0.4, w_2 = 0.1, w_3 = 0.1, w_4 = 0.4$ and $w_5 = 0.5, w_6 = 0.2, w_7 = 0.3$

Combination 18: $w_1 = 0.4, w_2 = 0.1, w_3 = 0.1, w_4 = 0.4$ and $w_5 = 0.7, w_6 = 0.1, w_7 = 0.2$

It is also important to note that we implement combination 1 where $w_1 = w_4 = 0.3, w_2 = w_3 = 0.2$ for equation (57) and $w_5 = w_7 = 0.4, w_6 = 0.2$ for equation (62) in our computational analysis since it has the most balanced weight values with the potential of reacting to all instance parameter (distances, loads, CS locations and ready dates) changes equivalently.

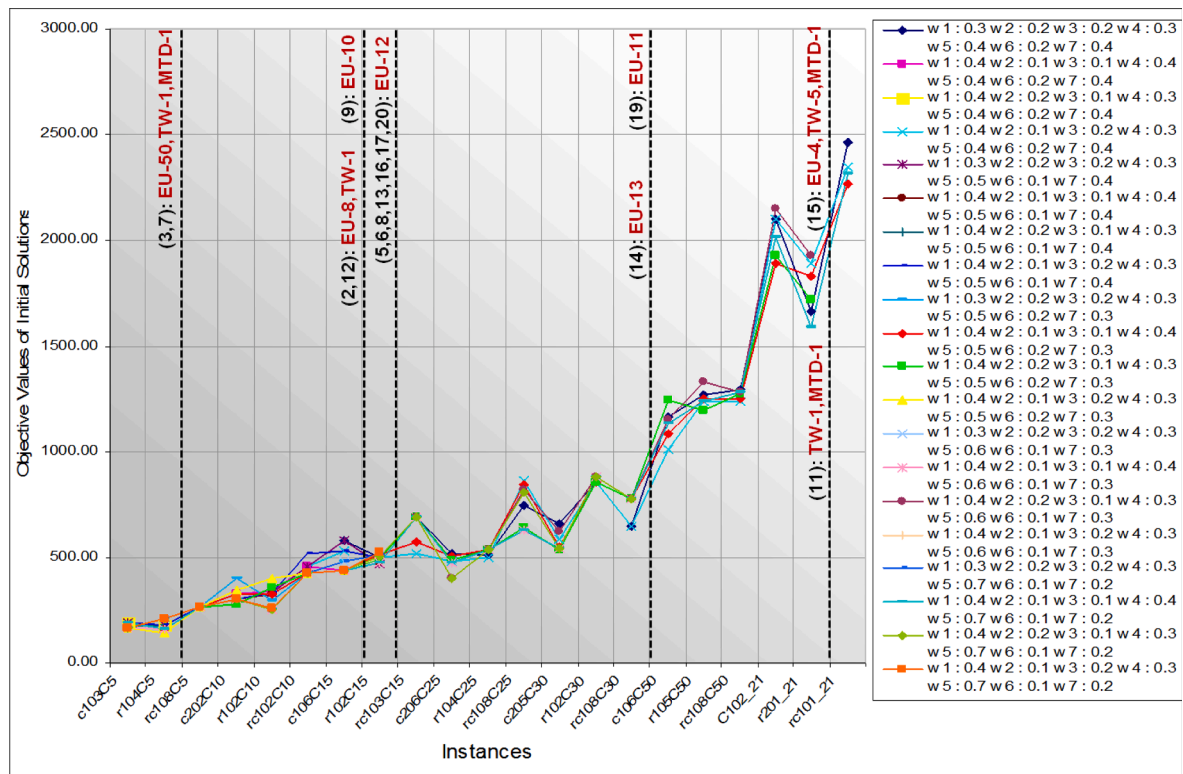


Fig. 4. Performance of weight combinations.

4.3.1.1. Effects of acceptance probability function and reheating mechanism. We also carry out a sensitivity analysis to investigate the effects of acceptance probability function and reheating mechanism on the performance of the proposed ALNS. Thus, we obtain new two versions of ALNS by removing the acceptance probability function from the ALNS, called $ALNS_{w_AF}$, and the reheating mechanism from the ALNS, called $ALNS_{w_RH}$. It should be noted that $ALNS_{w_AF}$ accepts a new solution whenever it improves the current one. All instances are solved by $ALNS_{w_AF}$ and $ALNS_{w_RH}$ and the best solutions among ten runs (Z^{w_AF} and Z^{w_RH}) are considered to determine the increase on total cost.

Tables 7–9 report computational results for small-, medium- and large-size instances, respectively. In the tables, while the first two columns stand for instance label and the best cost obtained by ALNS (Z^{ALNS}), the next two columns report the best cost obtained by $ALNS_{w_AF}$ and percentage increase on the cost ($\Delta_{w_AF} = 100 * ((Z^{w_AF} - Z^{ALNS}) / Z^{ALNS})$) and the last columns report the best cost obtained by $ALNS_{w_RH}$ and percentage increase on the cost ($\Delta_{w_RH} = 100 * ((Z^{w_RH} - Z^{ALNS}) / Z^{ALNS})$). It should be noted that the reheating mechanism has not already been used for small-size instances with 5 customers in the original ALNS. Thus, “*” in Table 7 indicates that sensitivity analysis is not performed for this group.

As it is seen from the tables, removing the acceptance probability function from the proposed ALNS causes a worsening on total cost averagely 7.19 %, 8.97 % and 4.74 % for small-, medium- and large-size instances, respectively. These results prove that the diversification effect of acceptance probability function improves the solution quality of the proposed ALNS. Similarly, removing the reheating mechanism increases total cost averagely 0.84 %, 2.53 % and 2.00 % for small-, medium- and large size- instances, respectively.

Hence, it is observed that the reheating improves the solution quality of the ALNS, though not as much as in the acceptance probability function.

Table 7
Effects of acceptance function and reheating mechanism on small-size instances.

Instance	Without acceptance function			Without reheating	
	Z^{ALNS}	Z^{w_AF}	Δ_{w_AF}	Z^{w_RH}	Δ_{w_RH}
c101C5	253.42	253.42	0.00	*	*
c103C5	168.37	168.37	0.00	*	*
c206C5	227.19	227.19	0.00	*	*
c208C5	175.35	175.35	0.00	*	*
r104C5	142.55	178.66	20.21	*	*
r105C5	157.14	158.45	0.83	*	*
r202C5	126.52	149.61	15.44	*	*
r203C5	198.54	198.54	0.00	*	*
rc105C5	238.73	243.81	2.08	*	*
rc108C5	262.43	262.43	0.00	*	*
rc204C5	184.11	186.06	1.05	*	*
rc208C5	179.62	193.14	7.00	*	*
Average			3.88		
c101C10	385.64	402.03	4.08	385.64	0.00
c104C10	259.92	293.65	11.49	283.92	8.46
c202C10	250.38	277.29	9.70	250.38	0.00
c205C10	237.24	331.53	28.44	237.24	0.00
r102C10	245.37	264.59	7.26	245.37	0.00
r103C10	196.22	196.22	0.00	196.22	0.00
r201C10	221.34	276.35	19.91	221.34	0.00
r203C10	213.65	241.67	11.60	213.65	0.00
rc102C10	397.08	418.82	5.19	397.08	0.00
rc108C10	347.15	356.58	2.64	347.15	0.00
rc201C10	320.22	364.76	12.21	339.29	5.62
rc205C10	334.45	348.66	4.08	346.98	3.61
Average			9.72		1.47
c103C15	361.27	389.64	7.28	370.59	2.52
c106C15	270.82	286.39	5.44	270.82	0.00
c202C15	374.91	463.76	19.16	376.92	0.53
c208C15	313.40	472.40	33.66	313.40	0.00
r102C15	387.98	393.55	1.42	387.98	0.00
r105C15	321.96	331.83	2.97	321.96	0.00
r202C15	380.58	398.51	4.50	377.04	−0.94
r209C15	280.09	280.09	0.00	280.09	0.00
rc103C15	406.33	412.40	1.47	406.33	0.00
rc108C15	385.10	405.55	5.04	386.22	0.29
rc202C15	420.89	422.31	0.34	420.89	0.00
rc204C15	329.14	384.47	14.39	329.14	0.00
Average			7.97		0.20

Table 8

Effects of acceptance function and reheating mechanism on medium-size instances.

Instance	Without acceptance function			Without reheating	
	Z^{ALNS}	Z^{W_AF}	Δ_{W_AF}	Z^{W_RH}	Δ_{W_RH}
c101C25	269.85	280.48	3.94	275.28	2.01
c103C25	331.31	375.99	13.49	335.20	1.17
c206C25	316.03	366.02	15.82	324.12	2.56
c208C25	277.84	314.78	13.30	291.04	4.75
r104C25	407.42	460.83	13.11	419.11	2.87
r105C25	539.87	539.87	0.00	544.89	0.93
r202C25	469.03	486.58	3.74	489.74	4.41
r203C25	410.28	429.34	4.64	417.66	1.80
rc105C25	559.33	634.34	13.41	566.05	1.20
rc108C25	447.55	595.63	33.09	449.43	0.42
rc204C25	372.53	409.42	9.90	374.86	0.63
rc208C25	349.84	402.87	15.16	370.73	5.97
Average			11.63		2.39
c101C30	327.86	375.68	12.73	328.00	0.04
c104C30	285.54	330.50	13.60	293.10	2.58
c202C30	360.07	389.52	7.56	383.40	6.08
c205C30	376.44	410.23	8.24	409.05	7.97
r102C30	668.16	676.14	1.18	670.03	0.28
r103C30	580.05	651.01	10.90	606.38	4.34
r201C30	614.55	661.81	7.14	619.11	0.74
r203C30	441.53	456.35	3.25	442.05	0.12
rc102C30	529.59	672.61	21.26	537.92	1.55
rc108C30	455.51	500.66	9.02	465.31	2.10
rc201C30	571.71	580.37	1.49	579.21	1.30
rc205C30	481.02	528.57	8.99	484.79	0.78
Average			8.78		2.32
c103C50	489.58	551.44	11.22	510.46	4.09
c106C50	611.62	691.68	11.57	653.44	6.40
c202C50	569.60	631.48	9.80	588.59	3.23
c208C50	533.33	580.64	8.15	544.91	2.12
r102C50	1033.55	1076.71	4.01	1058.40	2.35
r105C50	907.69	953.12	4.77	920.29	1.37
r202C50	629.93	674.60	6.62	650.99	3.24
r209C50	617.47	636.81	3.04	620.37	0.47
rc103C50	741.65	793.96	6.59	793.96	6.59
rc108C50	734.19	787.03	6.71	747.56	1.79
rc202C50	824.33	849.12	2.92	837.88	1.62
rc204C50	659.96	677.94	2.65	669.03	1.36
Average			6.50		2.88

Table 9

Effects of acceptance function and reheating mechanism on large-size instances.

Instance	Without acceptance function			Without reheating	
	Z^{ALNS}	Z^{W_AF}	Δ_{W_AF}	Z^{W_RH}	Δ_{W_RH}
C101_21	1090.60	1123.78	2.95	1088.57	-0.19
C102_21	1064.37	1106.63	3.82	1088.16	2.19
C201_21	1009.01	1014.05	0.50	1008.30	-0.07
C202_21	1018.55	1022.01	0.34	1033.18	1.42
r101_21	1623.14	1739.60	6.69	1696.33	4.31
r102_21	1486.91	1507.58	1.37	1479.64	-0.49
r201_21	1235.73	1253.36	1.41	1256.49	1.65
r202_21	1131.24	1225.06	7.66	1154.03	1.98
rc101_21	1714.47	1752.07	2.15	1714.13	-0.02
rc102_21	1547.10	1634.91	5.37	1680.09	7.92
rc201_21	1410.68	1553.64	9.20	1464.23	3.66
rc202_21	1337.54	1581.97	15.45	1360.71	1.70
Average			4.74		2.00

5. Conclusions

We have introduced and modelled a Mixed Fleet Vehicle Routing Problem with Time Windows and Partial Recharging by Multiple Chargers. This comprehensive problem holistically consolidates several aspects in the literature, such as mixed fleet, partial recharge policy, selection of recharging technology, emission limitation, emission and energy consumption functions depending on not only travelled distance but also load. We have developed an Adaptive Large Neighborhood Search (ALNS) based algorithm integrating innovative advanced neighborhood mechanisms to successfully handle complex problem constraints.

Furthermore, new approaches such as reheating, reinitialization and resetting the weights have been tailored for boosting diversification effect. New neighborhood scoring policy and a new enhancement procedure are also introduced. Selection of recharging technology among others at CS has been integrated in the solution phase of a mixed fleet problem. We have conducted extensive computational analyses to analyze the performance of our ALNS.

Our ALNS solved 30 instances (i.e. 28 small-size and 2 medium-size instances) to optimally. The average gap of the ALNS for 8 instances whose optimality are proven by CPLEX is 0.86 %. The ALNS yields feasible solutions for 28 instances (i.e. 16 medium-size and 12 large-size instances) for which CPLEX fails to find any solution within two hours. Upper bounds found by CPLEX for 45 instances (i.e. 2 for small-size, 31 for medium-size and 12 for large-size instances) are improved by the ALNS. The improvement ratios of upper bounds of CPLEX for small- and medium-size instances are 1.7 % and 19.55 %, respectively. Additionally, the ALNS outperforms CPLEX in terms of computation time by reaching optimal or near optimal solutions in very short times. Furthermore, average improvements of more than 18 % over initial solutions for all instances and average CVs of less than 2.5 for all instances showed that the proposed ALNS performs robustly. In conclusion, our ALNS successfully outperforms CPLEX on medium- and large-size instances in terms of both solution quality and computation time.

As a further research, the problem can be extended by considering heterogeneous ICVs and EVs. New neighborhood mechanisms could be proposed to reinforce algorithmic searching power. Exact solution algorithms based on branch-and-cut, branch-and-cut-and-price algorithms could be developed to obtain optimal solution or tight lower and upper bounds.

CRedit authorship contribution statement

Sercan Dönmez: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Software. **Çağrı Koç:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision. **Fulya Altıparmak:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tre.2022.102917>.

References

- Al-dal'ain, Çelebi, 2021. Planning a mixed fleet of electric and conventional vehicles for urban freight with routing and replacement considerations. *Sustainable Cities and Society*.
- Andelmin, J., Bartolini, E., 2017. An exact algorithm for the green vehicle routing problem. *Transportation Science* 51 (4), 1288–1303.
- Arslan, O., 2021. The location-or-routing problem. *Transportation Research Part B* 147, 1–21.
- Arslan, O., Yıldız, B., Karışan, O.E., 2015. Minimum cost path problem for plug-in hybrid electric vehicles. *Transportation Research Part E* 80, 123–141.
- Arslan, O., Karışan, O.E., Mahjoub, A.R., Yaman, H., 2019. A branch-and-cut algorithm for the alternative fuel refueling station location problem with routing. *Transportation Science* 53 (4), 1107–1125.
- Asghari, M., Mirzapour Al-e-hashem, S.M.J., 2021. Green vehicle routing problem: A state-of-the-art review. *International Journal of Production Economics* 231, 107899.
- Bruglieri, M., Mancini, S., Pisacane, O., 2019. The green vehicle routing problem with capacitated alternative fuel stations. *Computers & Operations Research* 112, 104759.
- Bruglieri, M., Ferone, D., Festa, P., Pisacane, O., 2022. A GRASP with penalty objective function for the green vehicle routing problem with private capacitated stations. *Computers & Operations Research* 143, 105770.
- Chen, Y., Li, D., Zhang, Z., Wahab, M.I.M., Jiang, Y., 2021. Solving the battery swap station location-routing problem with a mixed fleet of electric and conventional vehicles using a heuristic branch-and-price algorithm with an adaptive selection scheme. *Expert Systems with Applications* 186, 115683.
- Cortés-Murcia, D.L., Prodhon, C., Afsar, H.M., 2019. The electric vehicle routing problem with time windows, partial recharges and satellite customers. *Transportation Research Part E* 130, 184–206.
- Desaulniers, G., Errico, F., Irnich, S., Schneider, M., 2016. Exact algorithms for electric vehicle-routing problems with time windows. *Operations Research* 64 (6), 1388–1405.
- Dewi, S.K., Utama, D.M., 2021. A new hybrid whale optimization algorithm for green vehicle routing problem. *Systems Science & Control Engineering* 9 (1), 61–72.
- Erdelić, T., Carić, T., 2019. A survey on the electric vehicle routing problem: variants and solution approaches. *Journal of Advanced* 2019, 1–48.
- Erdogan, S., Miller-Hooks, E., 2012. A green vehicle routing problem. *Transportation Research Part E: Logistics and Transportation Review* 48 (1), 100–114.

- Felipe, Á., Ortuño, M.T., Righini, G., Tirado, G., 2014. A heuristic approach for the green vehicle routing problem with multiple technologies and partial recharges. *Transportation Research Part E* 71, 111–128.
- Figliozzi, M., 2010. Vehicle routing problem for emissions minimization. *Transportation Research Record* 2197 (1), 1–7.
- Ghorbani, E., Alinaghian, M., Gharehpetian, G., Mohammadi, S., Perboli, G., 2020. A survey on environmentally friendly vehicle routing problem and a proposal of its classification. *Sustainability* 12 (21), 9079.
- Goeke, D., Schneider, M., 2015. Routing a mixed fleet of electric and conventional vehicles. *European Journal of Operational Research* 245 (1), 81–99.
- Gonçalves, F., Cardoso, S.R., Relvas, S., Barbosa-Póvoa, A., 2011. Optimization of a distribution network using electric vehicles: A VRP problem. In: *15th Congresso Nacional Da Associação Portuguesa De Investigação Operacional*, pp. 18–20.
- Guo, F., Zhang, J., Huang, Z., Huang, W., 2022. Simultaneous charging station location-routing problem for electric vehicles: Effect of nonlinear partial charging and battery degradation. *Energy* 250, 123724.
- Hiermann, G., Hartl, R.F., Puchinger, J., Vidal, T., 2019. Routing a mix of conventional, plug-in hybrid, and electric vehicles. *European Journal of Operational Research* 272 (1), 235–248.
- Keskin, M., Çatay, B., 2016. Partial recharge strategies for the electric vehicle routing problem with time windows. *Transportation Research Part C* 65, 111–127.
- Keskin, M., Çatay, B., 2018. A matheuristic method for the electric vehicle routing problem with time windows and fast chargers. *Computers & Operations Research* 100, 172–188.
- Keskin, M., Çatay, B., Laporte, G., 2021. A simulation-based heuristic for the electric vehicle routing problem with time windows and stochastic waiting times at recharging stations. *Computers & Operations Research* 125, 105060.
- Koç, Ç., Karaoglan, I., 2016. The green vehicle routing problem: A heuristic based exact solution approach. *Applied Soft Computing* 39, 154–164.
- Koç, Ç., Jabali, O., Mendoza, J.E., Laporte, G., 2019. The electric vehicle routing problem with shared charging stations. *International Transactions in Operational Research* 26 (4), 1211–1243.
- Kucukoglu, I., Dewil, R., Cattrysse, D., 2021. The electric vehicle routing problem and its variations: A literature review. *Computers & Industrial Engineering* 161, 107650.
- Kumar, Ashwani, Kumar, Ravinder, Aggarwal, Ashutosh, 2022. A multi-objective route planning and charging slot reservation approach for electric vehicles considering state of traffic and charging station. *Journal of King Saud University-Computer and Information Sciences*.
- Lam, E., Desaulniers, G., Stuckey, P.J., 2022. Branch-and-cut-and-price for the Electric Vehicle Routing Problem with Time Windows, Piecewise-Linear Recharging and Capacitated Recharging Stations. *Computers & Operations Research* 145, 105870.
- Leggieri, V., Haouari, M., 2017. A practical solution approach for the green vehicle routing problem. *Transportation Research Part E: Logistics and Transportation Review* 104, 97–112.
- Li, J., Wang, D., Zhang, J., 2018. Heterogeneous fixed fleet vehicle routing problem based on fuel and carbon emissions. *Journal of Cleaner Production* 201, 896–908.
- Lin, J., Zhou, W., Wolfson, O., 2016. Electric vehicle routing problem. *Transportation Research Procedia* 12, 508–521.
- Macrina, G., Pugliese, L.D.P., Guerriero, F., Laporte, G., 2019a. The green mixed fleet vehicle routing problem with partial battery recharging and time windows. *Computers & Operations Research* 101, 183–199.
- Macrina, G., Laporte, G., Guerriero, F., Di Puglia Pugliese, L., 2019b. An energy-efficient green-vehicle routing problem with mixed vehicle fleet, partial battery recharging and time windows. *European Journal of Operational Research* 276 (3), 971–982.
- Marrekchi, E., Besbes, W., Dhoub, D., Demir, E., 2021. A review of recent advances in the operations research literature on the green routing problem and its variants. *Annals of Operations Research* 304 (1), 529–574.
- Montoya, A., Guéret, C., Mendoza, J.E., Villegas, J.G., 2016. A multi-space sampling heuristic for the green vehicle routing problem. *Transportation Research Part C: Emerging Technologies* 70, 113–128.
- Montoya, A., Guéret, C., Mendoza, J.E., Villegas, J.G., 2017. The electric vehicle routing problem with nonlinear charging function. *Transportation Research Part B* 103, 87–110.
- Olgun, B., Koç, Ç., Altıparmak, F., 2021. A hyper heuristic for the green vehicle routing problem with simultaneous pickup and delivery. *Computers & Industrial Engineering* 153, 107010.
- Paz, J.C., Granada-Echeverri, M., Escobar, J.W., 2018. The multi-depot electric vehicle location routing problem with time windows. *Int. J. Ind. Eng. Comput.* 9, 123–136.
- Pelletier, S., Jabali, O., Laporte, G., Veneroni, M., 2017. Battery degradation and behaviour for electric vehicles: Review and numerical analyses of several models. *Transportation Research Part B* 103, 158–187.
- Rabbani, M., Bosjin, S., Yazdanparast, R., Saravi, N., 2018. A stochastic time-dependent green capacitated vehicle routing and scheduling problem with time window, resiliency and reliability: a case study. *Decision Science Letters* 7 (4), 381–394.
- Rezgui, D., Siala, J.C., Aggoune-Mtalaa, W., Bouziri, H., 2019. Application of a variable neighborhood search algorithm to a fleet size and mix vehicle routing problem with electric modular vehicles. *Computers & Industrial Engineering* 130, 537–550.
- Ropke, S., Pisinger, D., 2006a. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation Science* 40, 455–472.
- Ropke, S., Pisinger, D., 2006b. A unified heuristic for a large class of vehicle routing problems with backhauls. *European Journal of Operational Research* 171, 750–775.
- Sadati, M.E.H., Çatay, B., 2021. A hybrid variable neighborhood search approach for the multi-depot green vehicle routing problem. *Transportation Research Part E* 149, 102293.
- Sassi, O., Cherif, W.R., Oulamar, A., 2014. *Vehicle routing problem with mixed feet of conventional and heterogenous electric vehicles and time dependent charging costs*. Technical Report. <https://hal.archives-ouvertes.fr/hal-01083966>.
- Schiffer, M., Walther, G., 2017. The electric location routing problem with time windows and partial recharging. *European Journal of Operational Research* 260, 995–1013.
- Schiffer, M., Walther, G., 2018. An adaptive large neighborhood search for the location-routing problem with intra-route facilities. *Transportation Science* 52, 331–352.
- Schneider, M., Stenger, A., Goeke, D., 2014. The electric vehicle routing problem with time windows and recharging stations. *Transportation Science* 48, 500–520.
- Solomon, M.M., 1987. Algorithms for the vehicle routing and scheduling problems with time window constraints. *Operations Research* 35, 254–265.
- Tahami, H., Rabadi, G., Haouari, M., 2020. Exact approaches for routing capacitated electric vehicles. *Transportation Research Part E* 144, 102126.
- Turkes, R., Sörensen, K., Hvattum, L.M., 2021. Meta-analysis of metaheuristics: Quantifying the effect of adaptiveness in adaptive large neighborhood search. *European Journal of Operational Research* 292, 423–442.
- Vidal, T., Laporte, G., Matl, P., 2020. A concise guide to existing and emerging vehicle routing problem variants. *European Journal of Operational Research* 286, 401–416.
- Vincent, F.Y., Jodiawan, P., Gunawan, A., 2021. An adaptive large neighborhood search for the green mixed fleet vehicle routing problem with realistic energy consumption and partial recharges. *Applied Soft Computing* 105, 107251.
- Xiao, Y., Zhang, Y., Kaku, I., Kang, R., Pan, X., 2021. Electric vehicle routing problem: A systematic review and a new comprehensive model with nonlinear energy recharging and consumption. *Renewable and Sustainable Energy Reviews* 151, 111567.
- Yang, J., Sun, H., 2015. Battery swap station location-routing problem with capacitated electric vehicles. *Computers & Operations Research* 55, 217–232.
- Zhang, J., Zhao, Y., Xue, W., Li, J., 2015. Vehicle routing problem with fuel consumption and carbon emission. *International Journal of Production Economics* 170, 234–242.